Highlights

Random Matrix Filtering and Planar Financial Networks for Volatility Forecasting in the Brazilian Stock Market

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- RMT separates market, group and noise components in B3 equities.
- PMFG filtering highlights localized sectoral dependency patterns.
- Within-sector correlations are larger than between-sector correlations on average.
- Ex-post network descriptors are associated with modest model-performance differences.
- Simple neural models provide an incremental deep-learning extension.

Random Matrix Filtering and Planar Financial Networks for Volatility Forecasting in the Brazilian Stock Market

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ABSTRACT

Financial markets are complex systems in which collective fluctuations, sectoral organization and sampling noise coexist in correlation matrices. We develop an econophysics workflow for Brazilian equities traded on B3 and conduct a retrospective test of whether the resulting dependency structure is associated with realized-volatility model performance. Using adjusted daily prices, we compute log returns for a 58-equity historical universe with raw coverage from 1998 to 2025. The RMT, clustering and network analyses use its complete synchronized panel, spanning 23 August 2006 to 15 August 2024, and are complemented by stylized-fact checks and chronological volatility-modelling experiments based on machine learning and simple deep-learning architectures. Using the Marčenko–Pastur benchmark, the spectrum is decomposed into a dominant market mode, candidate group/sector modes and a residual noise-compatible component; five eigenvalues lie above the Marčenko–Pastur upper bound, and the largest eigenvalue accounts for about 37% of the trace of the correlation matrix. Filtered MST and PMFG representations show that removing the market mode reduces average edge correlations while retaining localized sectoral patterns. The group-mode PMFG has lower mean edge correlation but higher clustering than the original network, suggesting stronger local clustering in the subsector-level representation. In a chronological train-validation-test evaluation, models augmented with ex-post RMT and network descriptors are competitive across selected horizons and metrics, but this does not establish real-time forecast gains because those structural descriptors are estimated from the full sample. Overall, RMT-filtered topology helps interpret market structure and motivates a future point-in-time volatility-risk design for emerging equity markets.

1. Introduction

Financial markets are often studied as complex systems in which heterogeneous agents, institutional constraints, liquidity cycles, information flows and feedback mechanisms interact across multiple time scales [1, 2]. Even when the analysis is restricted to daily asset returns, market variation is not fully described by the marginal behaviour of each series. It also has a relational component: how assets move together across the cross-section, summarized by their correlation matrix. This matrix reflects a mixture of collective market motion, sectoral organization, idiosyncratic shocks, crisis amplification and sampling noise. A central task is therefore to separate interpretable dependency structure from fluctuations that are compatible with random estimation effects [3].

The distinction matters because financial risk is inherently multivariate. The volatility of an individual asset matters, but portfolio risk, contagion, diversification and systemic vulnerability depend on how assets co-move [4]. During calm periods, correlations may appear moderate and dispersed. During stress periods, correlations often increase, reducing diversification benefits and lowering the effective dimensionality of the market [5, 6].

Emerging equity markets are useful in this context because they combine global risk channels with local institutional, macroeconomic and liquidity conditions. Brazil is a suitable setting: B3 (Brasil, Bolsa, Balcão), the country's

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stock exchange and financial-market infrastructure provider,[1] includes globally exposed commodity companies, large financial institutions, regulated utilities, consumer cyclicals, industrial firms, telecommunications firms and real-estate companies. These assets share exposure to domestic monetary policy, exchange-rate dynamics, country risk and global liquidity, but they also respond to distinct sectoral fundamentals. The dependency matrix therefore combines a dominant market component, group or sectoral components and a large residual component. Despite these features, the Brazilian equity market has received limited attention in the complex-systems literature. Previous studies have addressed Brazilian interbank networks [7], global financial system linkages [8], agrarian cross-correlations [9], multiscale networks [10] and equity-market topology [11], but a comprehensive B3 equity study combining Random Matrix Theory filtering, planar financial networks and volatility forecasting remains insufficiently developed.

Correlation matrices are noisy, especially when the number of assets is not negligible relative to the number of observations. Random Matrix Theory (RMT) offers a benchmark for this problem by comparing the eigenspectrum with the spectrum expected from random correlation matrices [12, 13, 14]. Eigenvalues lying inside the Marčenko–Pastur band are compatible with random sampling noise under the benchmark model, whereas eigenvalues outside the band indicate candidate collective modes that deserve further interpretation. Subsequent work has extended RMT-based methods for cleaning large covariance matrices [15]. We use RMT as an intermediate layer for reconstructing broad market, group-level and noise-compatible components before graph construction, not as a mechanical classifier of economic sectors.

Network filtering gives an additional representation of the same dependency system. By transforming correlations into distances, financial networks retain selected dependency links among assets and make it possible to inspect market backbones, local clusters and central nodes. The Minimum Spanning Tree (MST) emphasizes the sparsest connected structure [16], while the Planar Maximally Filtered Graph (PMFG) retains a denser planar representation with richer local connectivity [17, 18]. We use these constructions to ask whether strong dependencies are organized by sectors, whether central nodes change after market-mode removal and whether the intermediate-scale organization of the market contains information beyond average correlation. Filtered networks are also relevant for risk analysis, where peripheral assets can offer better diversification properties than central hubs [19, 20].

We develop an integrated econophysics workflow for Brazilian equities traded on B3. The workflow has three stages. First, we document stylized facts, static correlations, sectoral correlation differences and rolling market synchronization. Second, we apply RMT to decompose the correlation matrix into market, group/sector and noise components, followed by hierarchical clustering and ordered heatmaps. Third, we convert original and filtered matrices into MST and PMFG networks, study hub reorganization and aggregate dependencies at the subsector level. A chronological volatility-modelling experiment then examines whether ex-post structural descriptors are associated with machine-learning and simple deep-learning model performance. It is not presented as a fully point-in-time forecasting design because its RMT and network descriptors are estimated from the full sample.

The paper makes four contributions:

1. We build a reproducible econophysics workflow for B3 equities using adjusted daily prices with raw historical coverage from 1998 to 2025, including return construction, stylized-fact checks, correlation analysis, sectoral comparison and rolling synchronization.
2. We apply Random Matrix Theory to the complete 58-asset synchronized panel (23 August 2006–15 August 2024) and reconstruct market, group/sector and noise-compatible components of the empirical correlation matrix, yielding a spectral interpretation of Brazilian equity dependencies.
3. We compare MST and PMFG networks built from original and RMT-filtered matrices, with emphasis on hub reorganization, clique structure, clustering, sectoral retention and subsector-level aggregation.
4. We conduct a chronological, nested-feature-set evaluation of whether full-sample RMT and network descriptors are associated with realized-volatility model performance. This retrospective structural test motivates, but does not replace, a point-in-time forecasting design with recursively estimated features.

The article is organized as follows. Section 2 reviews financial dependencies, Random Matrix Theory, filtered financial networks and volatility forecasting. Section 3 describes the data and asset universe. Section 4 presents the methodology and the mathematical definitions used throughout the paper. Sections 5–10 report the results. Sections 11 and 12 discuss interpretation and limitations, and Section 13 presents the final considerations.

[1] https://www.b3.com.br/

2. Related work

2.1. Financial dependencies and econophysics

Econophysics studies financial markets using concepts and methods from statistical physics, nonlinear dynamics, complex systems and network science. A central observation is that financial returns exhibit stylized facts that are difficult to reconcile with independent Gaussian models. Heavy tails, volatility clustering, slow decay in absolute-return autocorrelations, intermittent bursts of activity and collective co-movement patterns are recurrent findings across asset classes, countries and sampling frequencies [1, 2]. These regularities motivate methods that focus not only on the marginal distribution of returns, but also on dependency, synchronization and network structure.

Financial dependency analysis lies at the intersection of econophysics and financial economics. In financial economics, dependence is central to portfolio risk, diversification, contagion, factor structure and systemic vulnerability. In econophysics, the same dependence is often represented through correlation matrices, eigenspectra, filtered graphs and hierarchical structures. Raddant and Di Matteo [3] emphasize that correlations, causality measures, volatility spillovers and networks offer additional views of financial systems. We follow this perspective: the goal is not merely to estimate pairwise correlations, but to decompose the correlation matrix, filter its noisy components, represent the remaining structure as networks and evaluate whether the extracted network structure has forecasting relevance.

Cross-correlation dynamics during crises have received attention because market stress often increases synchronization across assets. Zheng et al. [6] showed that changes in cross-correlation structure can serve as early indicators of systemic instability. Billio et al. [4] proposed econometric measures of connectedness and systemic risk in financial sectors. Junior and Franca [5] documented that correlations among global stock indices tend to increase during crises, reducing the effective dimensionality of international markets. These findings motivate the rolling-correlation and crisis-synchronization analysis undertaken in this paper, where market-wide co-movement is treated as an object of study instead of a nuisance.

The relevance of these studies for the present manuscript is twofold. First, they show that financial dependence is dynamic and crisis-sensitive. Second, they indicate that a single unfiltered correlation matrix may conceal different layers of structure: a broad market component, group or sectoral organization and residual noise. This motivates the use of Random Matrix Theory and filtered networks as additional tools for separating and interpreting these layers.

2.2. Random Matrix Theory in financial correlations

Random Matrix Theory entered finance as a way to separate information from noise in large correlation matrices. Laloux et al. [13, 21] showed that a substantial part of the eigenspectrum of financial correlation matrices can resemble the spectrum predicted for random matrices. The relevant benchmark is the Marčenko–Pastur distribution [12], which sets bounds for the eigenvalues of a random correlation matrix estimated from independent standardized time series. Plerou et al. [14] further studied eigenvalues and eigenvectors outside the random band and read the largest eigenvalue as a market-wide mode.

The intuition is that the empirical correlation matrix contains both structured and noisy components. If N assets are observed over T periods, the ratio $Q = T/N$ determines the width of the random-matrix noise band. Eigenvalues inside this band are treated as compatible with sampling noise, whereas eigenvalues above the upper edge are candidates for collective factors. This interpretation is useful in financial markets because broad market co-movement can dominate the first eigenvalue, while weaker sectoral or group modes may appear in subsequent eigenvectors.

Subsequent work extended RMT-based methods for covariance and correlation cleaning. Bun et al. [15] reviewed and unified several approaches for cleaning large covariance matrices using random matrix theory, connecting theoretical results with practical financial applications. Guhr et al. [22] give a broader mathematical review of random matrix theory and its applications in complex systems. In finance, RMT has been used not only for noise filtering, but also for portfolio optimization, factor identification and the study of collective market behaviour.

Our RMT interpretation is cautious. The largest eigenvalue captures a market mode because it reflects broad positive co-movement across many assets. Eigenvalues above the Marčenko–Pastur upper edge, excluding the first one, are read as group or sector candidates. Eigenvalues inside the random band are treated as noisy residual components. These labels should be read as filtering categories; they do not imply that each eigenvector has a unique economic interpretation. We use this spectral separation as the input for network construction and volatility forecasting in the Brazilian equity market.

2.3. Financial networks: MST, PMFG and filtered graphs

Financial networks transform a dependency matrix into a graph representation. The usual approach begins by converting correlations into distances, so that strongly correlated assets are close in the induced metric space. Mantegna [16] introduced the Minimum Spanning Tree (MST) as a way to reveal hierarchical organization in financial markets. The MST retains the connected tree with minimum total distance and therefore contains exactly $N - 1$ edges for N assets. Its main advantage is interpretability: it retains a sparse backbone of the strongest dependencies. Its main limitation is that it removes cycles and many potentially meaningful local connections.

Subsequent studies applied MST-based methods to study market taxonomy, instability and dynamics. Onnela et al. [23] analysed the evolution of MST-based market correlations and their implications for portfolio structure. Bonanno et al. [24] extended the network approach to multiple world markets, showing that MSTs often organize assets according to sectoral and geographical structure. Song et al. [25] studied the evolution of worldwide stock markets through correlation-based graphs, documenting time variation in network structure.

The Planar Maximally Filtered Graph (PMFG) relaxes the sparsity of the MST while preserving planarity [17]. For N nodes, a PMFG contains $3N - 6$ edges and includes the MST as a subgraph when both are constructed from the same ranked distance matrix. Because the PMFG can retain triangles and 4-cliques, it is more suitable for studying local connectivity, clustering, hub reorganization and group-level structure. Tumminello et al. [18] further explored the relationship between correlation, hierarchies and networks in financial markets. Massara et al. [26] later introduced the Triangulated Maximally Filtered Graph (TMFG), which extends this class of planar filtering methods and improves scalability for large systems.

Filtered networks are relevant for risk analysis. Pozzi et al. [19] used PMFG-based analysis to show that peripheral assets in financial networks can offer better diversification properties than central hubs. Aste et al. [20] studied correlation dynamics in volatile markets using filtered network approaches. Kenett et al. [27] used partial correlation analysis to reveal the dominant role of the financial sector in stock market networks. These studies motivate the hub-reorganization analysis presented here. By comparing original and RMT-filtered networks, we ask whether centrality is driven mainly by market-wide co-movement or whether localized central nodes emerge after removing the dominant market mode.

2.4. Financial networks in emerging markets

Network methods have also been applied to emerging markets, where market structure can differ substantially from that of large developed equity markets. Emerging markets may combine concentrated sectoral composition, macroeconomic sensitivity, lower liquidity in parts of the market and exposure to global risk cycles. These features make them attractive systems for dependency analysis because the correlation structure may reflect both domestic and international channels.

Tabak et al. [7] studied the structure of the Brazilian interbank network, documenting structural properties and vulnerability to targeted failures. Silva et al. [8] analysed the structure and dynamics of the global financial network, with attention to how emerging economies connect to the broader system. Wang et al. [28] studied the correlation structure of world stock markets using Pearson and partial-correlation networks, showing that emerging markets can display distinctive dependency patterns.

Studies focused more directly on Brazilian markets include Pereira et al. [10], who applied detrended cross-correlation analysis to construct multiscale networks of stock markets including B3, and Leonel Caetano and Yoneyama [11], who applied complex network analysis to the Brazilian stock market. Stosic et al. [9] studied correlations and cross-correlations in Brazilian agrarian commodities and stocks. These studies show that Brazilian financial data contain rich structural properties, but the literature remains fragmented across interbank networks, commodities, cross-market relations and equity-market structure.

The present paper contributes to this literature by focusing on a broad B3 equity universe and by connecting spectral filtering, planar networks and volatility forecasting in a single design. This integration matters because a network's interpretation depends on the matrix used to build it. A network constructed from the original correlation matrix may be dominated by the market mode, while a network constructed from group-mode components may reveal weaker but more localized dependency channels. This distinction is especially relevant in emerging markets, where common macroeconomic exposure and sectoral heterogeneity coexist.

2.5. Volatility forecasting and machine learning

Volatility forecasting has increasingly incorporated machine-learning methods because realized-volatility targets can be combined with large feature sets, nonlinear interactions and strict temporal validation. The theoretical foundations of realized-volatility measurement were established by Andersen et al. [29], and Poon and Granger [30] give an influential review of volatility forecasting methods. In the present study, the volatility-modelling component is retrospective: it asks whether features extracted ex post from RMT and filtered networks are associated with supervised-model performance for future realized volatility. It does not claim a point-in-time forecasting advantage from these full-sample structural descriptors.

Forecast evaluation is delicate because realized volatility is an observable proxy for latent conditional variance. Loss functions can rank models differently depending on their robustness to measurement error and their treatment of volatility underestimation. QLIKE is widely used because of its robustness properties under noisy volatility proxies [31]. For this reason, the forecasting experiment in this paper evaluates models using multiple metrics, including MAE, RMSE, QLIKE and out-of-sample R^2.

Machine-learning methods offer flexible nonlinear approximations and can incorporate large feature sets. Random Forests [32], gradient boosting [33] and regularized linear models [34] can combine lagged volatility, rolling statistics, market-wide variables and network centrality measures. Bucci [35] applied neural networks to realized-volatility forecasting, showing that flexible models can capture nonlinear volatility dynamics. Christensen et al. [36] developed a machine-learning framework for volatility forecasting and emphasized the importance of careful feature engineering. Zhang and Broadstock [37] explored how financial network structure relates to market risk from a machine-learning perspective.

In financial forecasting, however, flexibility can amplify noise and produce unstable gains if temporal validation is not strict. For this reason, we use a chronological train-validation-test split and treat network features as potential additional predictors, not replacements for lagged volatility information. The nested feature-set design follows an ablation logic: Set A measures the predictive value of standard volatility information, Set B tests the added value of market and RMT variables, and Set C tests whether MST and PMFG structure adds further information beyond these baseline predictors. Simple MLP and CNN-1D models are included as incremental neural extensions rather than as an exhaustive architecture search.

3. Data and asset universe

3.1. BovDB/B3 data source

The dataset consists of Brazilian equity prices from BovDB and its extended version BovDBv2 [38, 39]. BovDB was originally proposed as an open dataset of stock prices for companies traded on the Brazilian Stock Exchange (B3), covering the period from 1995 to 2020 and applying cumulative adjustment factors to correct prices for corporate events. BovDBv2 is distributed through Harvard Dataverse as the *Dataset for Intraday Analysis of B3 stock prices*, containing daily price information from 1995 to 2024 and intraday price data for Brazilian stocks from January 1995 to July 2024.

Although BovDBv2 includes intraday information, we use daily adjusted prices because the objective is to study long-run cross-sectional dependencies, Random Matrix Theory filtering and network structure over a broad historical window. Daily data give a stable sampling frequency for estimating correlations across many assets and reduce microstructure effects that would be more prominent in intraday analysis. The main objects of interest are adjusted closing prices, logarithmic returns and correlation matrices constructed from synchronized daily observations.

3.2. Adjusted prices

Adjusted closing prices are used because long historical equity samples are affected by stock splits, reverse splits, dividends, interest on equity, bonuses, corporate reorganizations and share-class changes. Without adjustment, these events can generate artificial jumps that contaminate return distributions, correlations and network structure. Return construction, standardization and correlation estimation are defined formally in Section 4.

3.3. Asset filters and equity universe definition

The core equity universe is obtained through deterministic filters applied to the BovDB/B3 daily database. The starting point is the set of B3 instruments with daily adjusted price information. We then exclude instruments that are not direct listed equities, including Brazilian Depositary Receipts (BDRs), exchange-traded funds (ETFs), investment funds, benchmark indices and other non-equity vehicles. This filter keeps the dependency matrix focused on listed equity shares instead of mixing equities with instruments that mechanically track baskets, benchmarks or foreign securities.

After this class filter, we retain observations only when adjusted closing prices are valid and strictly positive. Daily logarithmic returns are computed from adjusted closing prices, and extreme adjusted-return observations are inspected as part of the data-quality procedure. These observations are used as exclusion criteria only when they indicate clear corporate-action adjustment inconsistencies. Assets are then required to satisfy a minimum historical-coverage condition, and the remaining assets are synchronized on common trading dates. The formal notation for this synchronized return panel is given in Section 4.1.

3.4. Core historical universe

The core historical universe contains 58 assets with sufficient data coverage for the main RMT, clustering and network analysis. Its raw, unbalanced return file spans 17 March 1998 to 30 December 2025. Table 1 distinguishes that coverage from the complete synchronized analytical window. The universe is historical, not purely current. It allows us to study a long Brazilian equity sample while keeping the cross-section large enough for network analysis and small enough for interpretable visualizations.

Table 1: Summary of the samples used in this study.

Sample	Assets	Period	Main use
Representative assets	3	2006–2025	Stylized facts and initial forecasting experiment
Core historical universe	58	2006-08-23–2024-08-15	RMT, clustering and networks (complete panel)

The main spectral analysis uses a complete synchronized panel with $N = 58$ assets and $T = 1527$ common daily observations, from 23 August 2006 to 15 August 2024. The complete-case restriction is important because the empirical correlation matrix must be estimated from a common set of trading dates across all assets. This avoids pairwise changes in sample composition and ensures that eigenvalues, eigenvectors, distances and network edges are derived from a single common correlation matrix. Static pairwise correlations and rolling correlations are reported separately using the observations available for each pair or window; they therefore retain the broader unbalanced historical coverage and should not be interpreted as estimates from the complete RMT panel.

3.5. Demonstration and forecasting assets

We use PETR4, VALE3 and BBDC4 as representative assets for stylized facts and as the initial forecasting targets. They were selected because they are large, liquid and economically distinct Brazilian equities. PETR4 represents oil and gas exposure, with sensitivity to commodity prices and firm-specific institutional events. VALE3 represents basic materials and global mining exposure, with sensitivity to commodity cycles and exchange-rate dynamics. BBDC4 represents the financial sector and domestic credit conditions.

This small panel is not intended to represent the entire Brazilian equity market. It offers a compact and interpretable entry point into the analysis, illustrating heavy tails, volatility clustering, return extremes, absolute-return persistence and the feasibility of the forecasting design.

3.6. Sector and subsector classification

Each asset in the core historical universe is assigned to a macro-sector, subsector and segment classification. The macro sectors include Financials, Oil Gas and Biofuels, Basic Materials, Utilities, Industrials, Consumer Cyclical, Consumer Non-Cyclical, Real Estate and Telecommunications. These labels are used for descriptive comparison, sectoral correlation tests, heatmap interpretation and subsector-level aggregation.

Utilities and Real Estate are kept separate because their risk drivers differ substantially. Utilities are more directly linked to regulation, concessions, inflation adjustment mechanisms and defensive cash flows. Real Estate is more sensitive to credit conditions, interest rates and the real-estate cycle. Within Basic Materials, the subsector classification distinguishes Mining, Steel and Metallurgy, Chemicals, and Paper and Cellulose. This distinction matters in Brazil because commodity-linked firms can share global exposure while still belonging to different industrial chains.

3.7. Data quality notes and limitations

The main data-quality issue is the distinction between genuine extreme returns and artificial jumps caused by corporate-action adjustment problems. Adjusted prices reduce the impact of splits, dividends and related events, but they do not guarantee that every historical corporate action is perfectly corrected. This limitation is especially relevant in long emerging-market samples, where ticker histories, share-class changes and corporate reorganizations can be complex.

For PETR4, VALE3 and BBDC4, the largest daily returns observed in the 2006–2025 window match stress periods such as the 2008 global financial crisis, the Brazilian recession period around 2015–2016 and the 2020 COVID shock. Extreme-return validation is treated as a separate quality-control step, and assets with implausible corporate-action-related jumps are excluded from the main analyses.

The main RMT, clustering and network results are based on the complete synchronized panel within the core historical universe. This choice yields a common panel for estimating the corresponding correlation matrix, eigenvalue spectrum and network structure, but it also imposes limits: the complete-panel window is shorter than the raw historical coverage. Results may differ under intraday sampling, alternative liquidity filters, rolling-window universes, or broader target sets for forecasting. These extensions are left for robustness analysis and future work.

4. Methodology

The methodology transforms adjusted B3 prices into return matrices, correlation matrices, spectral components, filtered financial networks and volatility-modelling features. Figure 1 summarizes the workflow. The workflow estimates dependencies and separates market-wide motion, group-level structure and noise-compatible components before conducting a retrospective evaluation of whether the resulting network structure is associated with model performance.

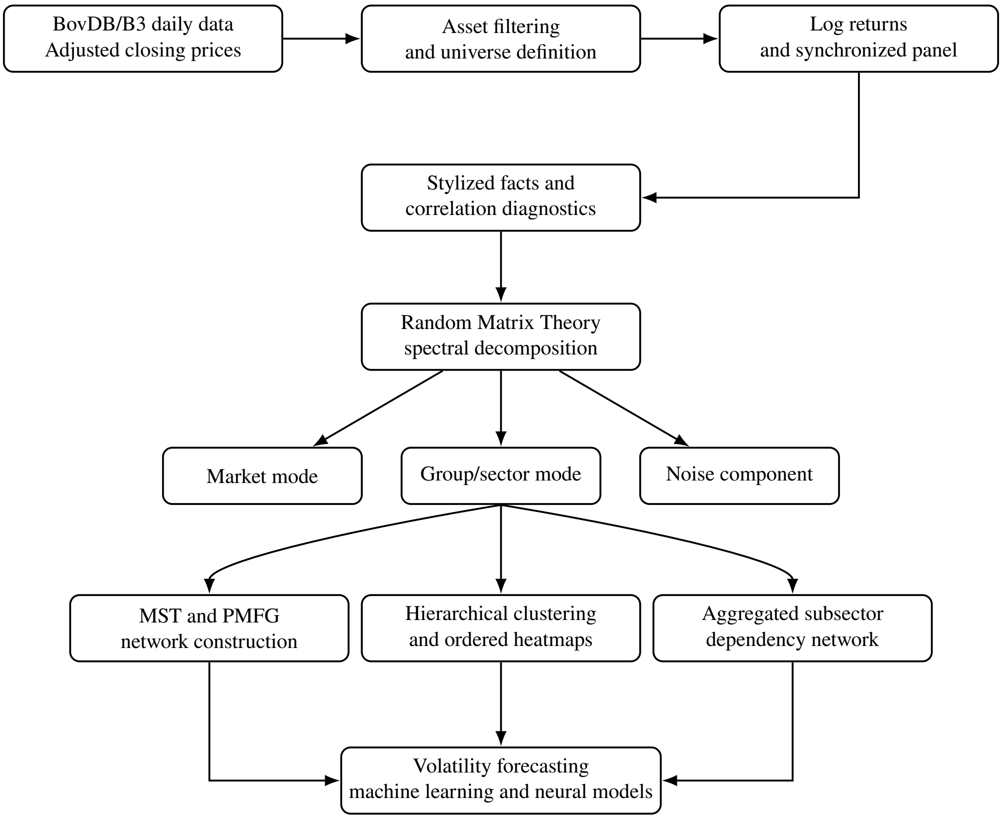


Figure 1: Overview of the workflow. The workflow proceeds through stylized-fact checks, correlation estimation, Random Matrix Theory filtering, hierarchical clustering, MST/PMFG network construction, subsector aggregation and volatility forecasting with machine-learning, deep-learning, RMT and network-based features.

4.1. Notation and return matrix

We fix notation before defining the main steps. The index $i = 1, \dots, N$ identifies assets, and the index $t = 1, \dots, T$ identifies trading days in a synchronized panel. In the main spectral, clustering and network analysis, $N = 58$ is the number of assets and $T = 1527$ is the number of common daily observations after synchronization. Thus T is the number of dates in the complete panel, not the total number of scalar return entries.

Let $P^{\text{adj}}_{i,t}$ denote the adjusted closing price of asset i on trading day t. Daily logarithmic returns are computed as

$$r_{i,t} = \log P^{\text{adj}}_{i,t} - \log P^{\text{adj}}_{i,t-1}, \tag{1}$$

where $P^{\text{adj}}_{i,t} > 0$ and $P^{\text{adj}}_{i,t-1} > 0$. Log returns are used because they are additive across time and standard in studies of financial returns.

The return panel is represented by the matrix

$$\mathbf{R} = \left(r_{i,t}\right)_{t=1,\dots,T;\, i=1,\dots,N} \in \mathbb{R}^{T\times N}. \tag{2}$$

For correlation-based analysis, each return series is standardized within the relevant sample:

$$\tilde{r}_{i,t} = \frac{r_{i,t} - \mu_i}{\sigma_i}, \tag{3}$$

where μ_i and σ_i are the sample mean and standard deviation of asset i. The standardized return matrix is denoted by $\widetilde{\mathbf{R}}$.

Let $\mathbf{C}$ denote the Pearson correlation matrix, and let C_{ij} be the Pearson correlation between assets i and j. In the Random Matrix Theory step, λ_k denotes the k-th eigenvalue of $\mathbf{C}$, $\mathbf{u}_k$ denotes its normalized eigenvector and $Q = T/N$ denotes the sample-size-to-dimension ratio used in the Marčenko–Pastur benchmark.

4.2. Asset filtering and complete return panel

Let $\mathcal{A}_0$ denote the initial set of B3 instruments available in the BovDB/B3 daily database. The data section explains the economic motivation for excluding non-direct-equity instruments. Formally, let $\mathcal{X}$ be the excluded set containing BDRs, ETFs, investment funds, benchmark indices and other non-equity vehicles. The eligible direct-equity set is

$$\mathcal{A}_1 = \mathcal{A}_0 \setminus \mathcal{X}, \tag{4}$$

where each element of $\mathcal{A}_1$ is an asset identifier, not an observation date.

For each remaining asset, observations are retained only when adjusted closing prices are valid:

$$P^{\text{adj}}_{i,t} > 0. \tag{5}$$

Daily log returns are then computed from adjusted prices. Assets with insufficient historical coverage are removed according to

$$T_i \geq T_{\min}, \tag{6}$$

where T_i is the number of valid return observations for asset i and $T_{\min}$ is the minimum coverage threshold required for the target analysis.

After filtering, assets are synchronized by common trading dates. The RMT, clustering and network analyses are performed on a complete return panel

$$\mathbf{R}_{\text{core}} \in \mathbb{R}^{T\times N}, \tag{7}$$

with $N = 58$ assets and $T = 1527$ complete daily observations in the main spectral analysis. Complete-panel estimation ensures that all eigenvalues, eigenvectors, distances and network edges are derived from a single common correlation matrix estimated from assets observed on common trading dates.

4.3. Stylized facts and descriptive diagnostics

We first document stylized facts for representative B3 assets. For each asset, we compute the sample mean, standard deviation, skewness, excess kurtosis, minimum and maximum returns, value-at-risk at 1% and 5%, counts of large absolute returns and autocorrelations of absolute returns.

For an asset return series $\{r_t\}$, the autocorrelation of absolute returns at lag ℓ is

$$\rho_{|r|}(\ell) = \mathrm{Corr}(|r_t|, |r_{t-\ell}|). \tag{8}$$

These checks assess whether B3 returns display the regularities commonly associated with financial time series: heavy tails, volatility clustering and persistent volatility [2].

4.4. Correlation matrix estimation

The Pearson correlation matrix is estimated from standardized returns as

$$C_{ij} = \frac{1}{T-1} \sum_{t=1}^{T} \tilde{r}_{i,t}\tilde{r}_{j,t}. \tag{9}$$

Equivalently, in matrix form,

$$\mathbf{C} = \frac{1}{T-1} \widetilde{\mathbf{R}}^{\top}\widetilde{\mathbf{R}}. \tag{10}$$

The matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$ is symmetric and has unit diagonal. For pairwise descriptive analysis, correlations may be computed with available observations for each pair. For RMT, clustering, MST and PMFG construction, the complete synchronized panel is used to obtain a single consistent $N \times N$ matrix.

4.5. Sectoral correlation comparison

Each asset is assigned to a macro-sector and subsector. The sectoral comparison asks a simple question: are correlations between assets in the same macro-sector larger than correlations between assets in different macro-sectors? To answer this, we classify every unordered asset pair (i, j) into one of two groups. The indicator s_{ij} is not a correlation coefficient; it only records whether the two assets belong to the same macro-sector:

$$s_{ij} = \begin{cases} 1, & \text{if assets } i \text{ and } j \text{ belong to the same macro-sector,} \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

After this classification, the corresponding Pearson correlations C_{ij} are split into two samples. Pairs with $s_{ij} = 1$ form the within-sector correlation distribution, while pairs with $s_{ij} = 0$ form the between-sector correlation distribution. We report their mean, median and dispersion, and use the difference in means

$$\Delta_{\text{sector}} = \overline{C}_{\text{within}} - \overline{C}_{\text{between}} \tag{12}$$

as the test statistic. Its significance is assessed with a one-sided asset-level sector-label permutation test. In each of 10,000 fixed-seed permutations, the macro-sector labels are reassigned across the 58 assets while preserving the sector-size vector; the observed correlation matrix and its overlapping-pair dependence are left unchanged. The Monte Carlo p-value is the fraction of permuted statistics at least as large as the observed statistic, with the standard plus-one correction. This avoids treating the 1,653 overlapping pairwise correlations as independent observations. The test is conditional on the initial classification and on exchangeability of sector labels under the null. It evaluates whether the economic sector classification is reflected in the dependency matrix before any RMT filtering is applied.

4.6. Rolling market synchronization

Static full-sample correlations hide time variation in market dependence. To measure market synchronization, rolling pairwise correlations are estimated over a window of length w. For each window ending at time t, the average market correlation is

$$\bar{\rho}_t^{(w)} = \frac{2}{N(N-1)} \sum_{1\le i<j\le N} \rho_{ij,t}^{(w)}, \tag{13}$$

where $\rho_{ij,t}^{(w)}$ is the Pearson correlation between assets i and j computed within the rolling window. Elevated values of $\bar{\rho}_t^{(w)}$ indicate stronger market-wide synchronization and motivate the subsequent spectral decomposition.

4.7. Random Matrix Theory filtering

Random Matrix Theory is used to distinguish structured collective modes from eigenvalue fluctuations compatible with sampling noise. Let the spectral decomposition of the empirical correlation matrix be

$$\mathbf{C} = \sum_{k=1}^{N} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top, \tag{14}$$

where λ_k is the k-th eigenvalue and $\mathbf{u}_k$ is the corresponding normalized eigenvector. Eigenvalues are sorted in decreasing order, so $k = 1$ denotes the largest mode.

For a random correlation matrix estimated from N independent standardized time series of length T, the Marčenko–Pastur bounds are

$$\lambda_\pm = \sigma^2 \left(1 + \frac{1}{Q} \pm 2\sqrt{\frac{1}{Q}}\right), \qquad Q = \frac{T}{N}. \tag{15}$$

Since standardized returns are used, $\sigma^2 = 1$. Eigenvalues inside $[\lambda_-, \lambda_+]$ are treated as compatible with random-matrix noise, while eigenvalues above λ_+ are candidates for collective modes.

Let K denote the number of sample eigenvalues above λ_+. The dominant market component is defined as

$$\mathbf{C}^{\text{market}} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top. \tag{16}$$

The group or sector component is reconstructed from the remaining retained eigenvalues:

$$\mathbf{C}^{\text{group}} = \sum_{k=2}^{K} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top. \tag{17}$$

The retained-mode matrix is

$$\mathbf{C}^{\text{filtered}} = \sum_{k=1}^{K} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top, \tag{18}$$

and the residual noisy component is

$$\mathbf{C}^{\text{noise}} = \sum_{k=K+1}^{N} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top. \tag{19}$$

As an issuer-level robustness check, we repeat the RMT calculation after collapsing multiple listed share classes into one representative series per company. For each company, the representative is the ticker with the largest number of valid daily returns in the raw historical file; ties are resolved alphabetically. This deterministic rule removes mechanical within-issuer duplication while avoiding a liquidity-based selection rule that is unavailable uniformly across the long sample. The same complete-case correlation and Marčenko–Pastur calculations are then applied to the resulting issuer-collapsed panel.

We also assess the five leading eigenvalues against a nonparametric circular-shift null. For each of 1,000 fixed-seed surrogate panels, each asset-return series is shifted by an independently sampled circular offset. This preserves its marginal values and temporal ordering while breaking contemporaneous alignment across assets. The empirical eigenvalue at each rank is compared with the corresponding rank-specific surrogate distribution. This test complements the idealized Marčenko–Pastur benchmark; it is a test of synchronized dependence, not a sector-classification rule.

Note that partial spectral reconstructions generally do not preserve the unit diagonal of the original correlation matrix. When these filtered matrices are used for downstream tasks, they are rescaled to unit diagonal if the subsequent method requires valid Pearson correlation bounds. This decomposition separates networks dominated by broad market co-movement from networks that emphasize localized group and sectoral structure.

4.8. Mantegna distance and hierarchical clustering

Correlations are converted into distances using the metric introduced by Mantegna [16]:

$$d_{ij} = \sqrt{2(1 - C_{ij})}. \tag{20}$$

For a valid correlation matrix, this transformation maps strongly positively correlated assets to short distances and less positively correlated or negatively correlated assets to longer distances. The resulting distance matrix is used for hierarchical clustering, ordered heatmaps, MST and PMFG construction.

Hierarchical clustering is applied to the distance matrix using average linkage (UPGMA). The cophenetic correlation [40] is used to assess how faithfully the dendrogram preserves pairwise distances. Ordered heatmaps use the dendrogram ordering to rearrange the correlation matrices, making block structures visually identifiable. Comparing original, filtered and group-mode heatmaps helps distinguish broad market co-movement from localized sectoral dependency.

4.9. Minimum Spanning Tree and Planar Maximally Filtered Graph

A financial network is represented as

$$G = (V, E, w), \tag{21}$$

where V is the set of assets, E is the retained set of edges and $w_{ij} = C_{ij}$ is the dependency weight associated with edge (i, j).

The Minimum Spanning Tree is the connected acyclic graph that minimizes total distance:

$$\mathrm{MST} = \arg\min_{E'} \sum_{(i,j)\in E'} d_{ij}, \tag{22}$$

subject to (V, E') being connected and acyclic. For N nodes, the MST has $N - 1$ edges. With $N = 58$, this gives 57 edges.

The Planar Maximally Filtered Graph is constructed by considering candidate edges in increasing order of distance, retaining them if they do not violate planarity. For N nodes, a PMFG has

$$|E_{\mathrm{PMFG}}| = 3N - 6 \tag{23}$$

edges, which gives 168 edges for $N = 58$. Unlike the MST, the PMFG can contain cycles, triangles and 4-cliques. Under the same ranking of pairwise distances, the MST is contained in the PMFG as a subgraph. This makes the PMFG a less aggressive graph filter that preserves richer local geometry.

For each network, we report edge correlations, same-sector and same-subsector edge ratios, average clustering, node degree, betweenness centrality and hub rankings. Comparing original and group-mode networks identifies whether centrality is driven by broad market co-movement or by localized dependency channels. We assess temporal stability by splitting the complete 58-asset panel into three consecutive equal-sized windows of 509 trading days and re-estimating the correlation matrix, RMT group component, MST and PMFG in each window. For each network/matrix combination, we report pairwise edge Jaccard overlap, overlap of the five largest betweenness hubs, and adjusted Rand index (ARI) between unweighted greedy-modularity community assignments. These are descriptive stability diagnostics, not independent statistical tests; their role is to identify which graph features persist across windows.

4.10. Aggregated subsector dependency network

The asset-level networks are complemented by an aggregated subsector dependency network. Let $g(i)$ denote the subsector associated with asset i. For two subsectors a and b, the aggregated dependency weight is defined as

$$W_{ab} = \frac{1}{|\mathcal{E}_{ab}|} \sum_{(i,j)\in\mathcal{E}_{ab}} w_{ij}, \tag{24}$$

where

$$\mathcal{E}_{ab} = \{(i, j) \in E : g(i) = a,\ g(j) = b\}. \tag{25}$$

The aggregation converts ticker-level dependencies into a subsector-level map of dependency channels. The resulting network is summarized using edge density, average dependency, weighted degree and betweenness centrality.

4.11. Volatility forecasting formulation

The ideal forecasting task is formulated as a supervised time-series prediction problem. For target asset i and horizon h, realized volatility is defined as

$$RV_{i,t,h} = \sqrt{\sum_{\tau=1}^{h} r_{i,t+\tau}^2}. \tag{26}$$

In a strictly point-in-time implementation, the objective would be to predict $RV_{i,t,h}$ using an information set $\mathcal{F}_t$ available at time t:

$$\widehat{RV}_{i,t,h} = f_\theta(\mathcal{F}_t). \tag{27}$$

For evaluation under metrics such as QLIKE, these realized volatility and predicted volatility measures are squared to represent realized variance proxies. The target horizons are $h = 5$ and $h = 20$ trading days. The initial experiment uses PETR4, VALE3 and BBDC4, with chronological training, validation and test periods. This preserves the ordering of model fitting and selection, but the experiment is not a strict implementation of $\mathcal{F}_t$: the RMT and network descriptors are estimated once from the full synchronized panel and then attached to observations. The resulting comparison is therefore retrospective and structural, rather than a real-time forecasting test.

4.12. Machine-learning and deep-learning feature sets

Three feature sets are used in a nested ablation design. Set A contains classical return and volatility predictors, including rolling volatilities, rolling absolute returns, lagged realized volatility, EWMA volatility and higher-moment rolling statistics. Set B adds rolling market variables and full-sample RMT descriptors such as the largest eigenvalue, market-mode trace share and number of eigenvalues above the random-matrix bound. Set C adds full-sample MST and PMFG descriptors, including degree, betweenness and clustering measures from original and group-mode networks.

This nesting defines a controlled ablation:

$$\mathcal{F}_t^A \subset \mathcal{F}_t^B \subset \mathcal{F}_t^C. \tag{28}$$

Set A provides the point-in-time volatility baseline. Sets B and C test whether ex-post market, RMT and network descriptors are associated with changes in model performance beyond that baseline. Ridge regression, Random Forest and histogram gradient boosting are used to evaluate these increasingly rich information sets [34, 32, 33]. A strict real-time extension would reconstruct the RMT and network quantities recursively within each training window, using no observations after the forecast origin.

The deep-learning extension is intentionally incremental. A multilayer perceptron (MLP) is trained on the same tabular feature sets, using positive-output regression for realized volatility. A one-dimensional convolutional neural network (CNN-1D) is trained on short rolling windows of return and volatility features. These neural models are included to test whether a simple nonlinear representation adds forecasting value; they are not presented as an exhaustive search over neural architectures [35, 36].

4.13. Forecast evaluation metrics

Forecasts are evaluated using MAE, RMSE, out-of-sample R^2 and QLIKE. For realized values y_t and forecasts $\hat{y}_t$, the MAE and RMSE are

$$\text{MAE} = \frac{1}{M}\sum_{t=1}^{M} |y_t - \hat{y}_t|, \tag{29}$$

and

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{t=1}^{M}(y_t - \hat{y}_t)^2}. \tag{30}$$

The out-of-sample coefficient of determination is

$$R^2_{\text{oos}} = 1 - \frac{\sum_{t=1}^{M}(y_t - \hat{y}_t)^2}{\sum_{t=1}^{M}(y_t - \bar{y}_{\text{train}})^2}. \tag{31}$$

The QLIKE loss is

$$\mathrm{QLIKE}_t = \log(\hat{\sigma}_t^2) + \frac{\sigma_t^2}{\hat{\sigma}_t^2}, \tag{32}$$

where $\hat{\sigma}_t^2$ is the predicted variance proxy and σ_t^2 is the realized variance proxy. QLIKE is included because it is widely used for volatility forecast comparison when realized volatility is an imperfect proxy for latent variance [31]. To quantify uncertainty in test-period QLIKE, we use 2,000 fixed-seed circular moving-block bootstrap replications, with block length equal to the forecast horizon. This retains the serial dependence induced by overlapping realized-volatility targets. Aggregate intervals resample the date-level mean loss across the three target assets, preserving their contemporaneous co-movement. As a focused formal contrast, the validation-selected best structural specification is compared with the validation-selected Set A specification using the date-level loss differential $d_t = \mathrm{QLIKE}_t^A - \mathrm{QLIKE}_t^{\mathrm{struct}}$. A one-sided HAC test uses a Bartlett/Newey–West long-run variance with lag $h-1$. Positive d_t favours the structural specification. This inferential check remains retrospective because the structural descriptors are ex-post.

5. Stylized facts and correlation structure

5.1. Stylized facts of selected B3 assets

Figure 2 summarises the return properties of PETR4, VALE3 and BBDC4. We use these assets as representative examples and as the initial forecasting targets because they are liquid, economically important and sectorally distinct; they are not intended to be a proxy for the entire Brazilian equity market. The figure combines four checks that are standard in the econophysics literature [2]: normalized prices, daily returns, the complementary cumulative distribution of absolute returns and the autocorrelation of absolute returns.

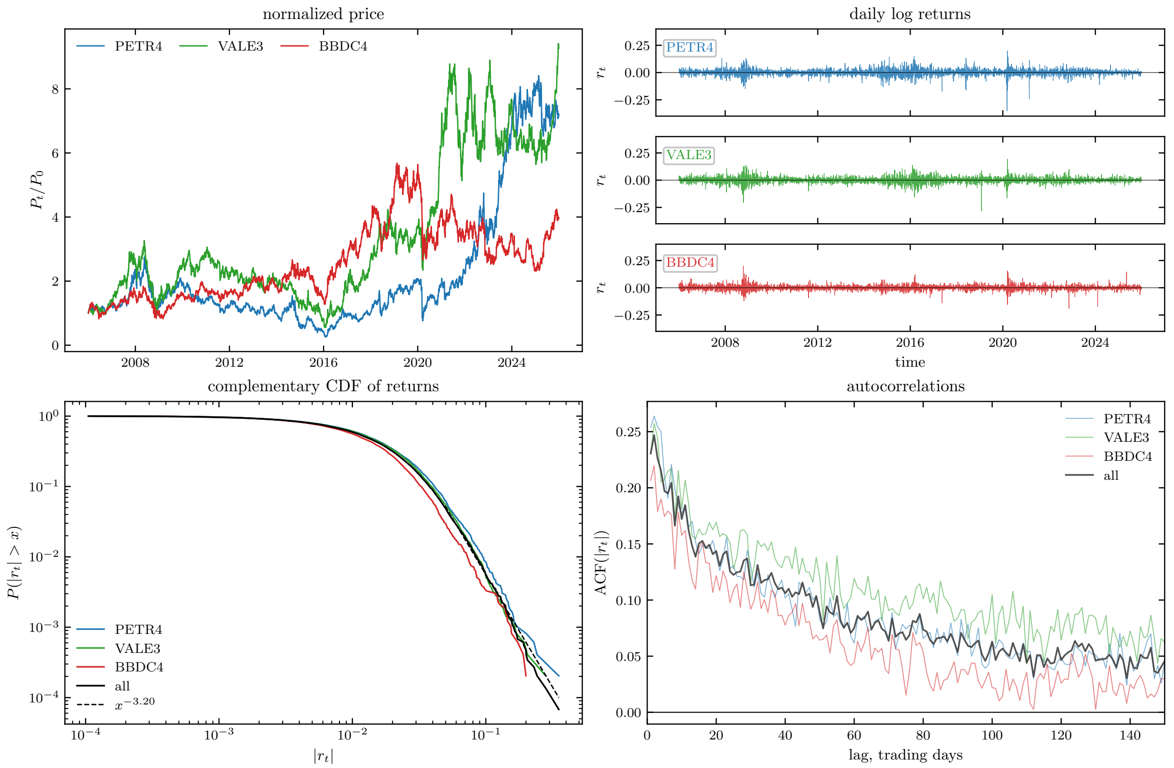


Figure 2: Stylized facts for PETR4, VALE3 and BBDC4 over 2006–2025. The panels show normalized prices, daily log returns, the complementary cumulative distribution of absolute returns and autocorrelations of absolute returns. The return panel uses three vertically aligned subpanels with a common scale and a zero reference line for each asset.

The normalized-price panel illustrates heterogeneous long-run trajectories across the three assets. This heterogeneity is expected because the selected firms belong to distinct economic segments and are exposed to different sectoral and macro-financial drivers. The return panel displays periods in which large movements cluster over time, a pattern indicating volatility clustering. The CCDF panel suggests that large absolute returns occur more frequently than would be expected under a Gaussian benchmark, pointing to heavy-tailed returns. The autocorrelation panel shows positive dependence in $|r_t|$ over several lags, indicating persistent volatility [2].

5.2. Descriptive statistics

Table 2 quantifies the same diagnostics. All three series have means close to zero, substantial kurtosis and non-negligible counts of large absolute returns. PETR4 has the largest negative daily observation in the 2006–2025 window, while VALE3 and BBDC4 also show stress-period extremes. The autocorrelations of absolute returns remain positive across several lags, especially for VALE3.

The descriptive statistics match the visual reading of Figure 2. The return distributions display substantial departures from a Gaussian benchmark, including excess kurtosis and persistent absolute-return autocorrelations. These properties match the well-documented stylized facts of financial returns [2] and support treating the Brazilian market as a complex system in which dependency structure, not marginal normality, is the relevant modelling object.

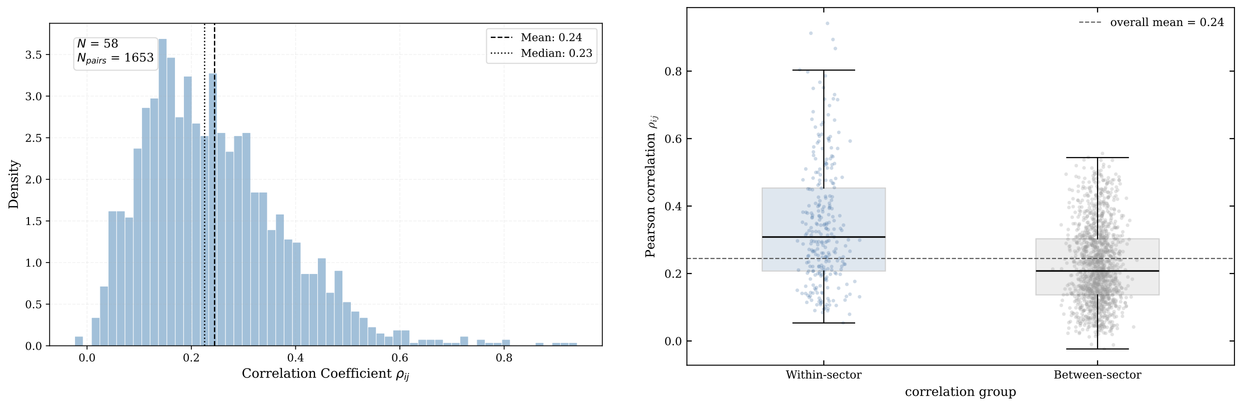
Table 2: Descriptive statistics for the three demonstration assets, 2006–2025.

Symbol	n	Mean	Std.	Skew.	Ex. kurt.	Min.	Max.	$\mathrm{VaR}_{1\%}$	$\mathrm{VaR}_{5\%}$	$\lvert r\rvert > 10\%$	$\lvert r\rvert > 20\%$	ACF_1	ACF_5	ACF_{20}	ACF_{60}
BBDC4	4953	0.00028	0.02180	0.02	7.96	-0.1910	0.1999	-0.0559	-0.0312	16	0	0.207	0.174	0.122	0.044
PETR4	4953	0.00041	0.02755	-0.76	11.55	-0.3524	0.2007	-0.0767	-0.0408	41	4	0.254	0.209	0.140	0.062
VALE3	4953	0.00045	0.02576	-0.22	7.51	-0.2814	0.1936	-0.0676	-0.0381	26	2	0.231	0.207	0.169	0.119

Notes: Mean and Std. are the sample mean and standard deviation of daily log returns. Skew. is sample skewness and Ex. kurt. is excess kurtosis relative to the Gaussian benchmark. $\mathrm{VaR}_{1\%}$ and $\mathrm{VaR}_{5\%}$ are the empirical 1% and 5% lower-tail return quantiles. $|r| > 10\%$ and $|r| > 20\%$ count daily observations whose absolute log return exceeds the stated threshold. ACF_ℓ is the autocorrelation of absolute returns at lag ℓ trading days.

5.3. Static correlation distribution

We now turn from individual return behaviour to cross-sectional dependence. The core historical universe contains 58 assets, producing $N(N-1)/2 = 1653$ unique asset pairs. The static pairwise summaries below use the available overlap for each pair and therefore describe the broader unbalanced historical coverage; Figure 3 summarizes their distribution and then splits pairs into within-sector and between-sector groups. The subsequent RMT decomposition instead uses the single complete 58-asset panel from 23 August 2006 to 15 August 2024.



(a) Full pairwise distribution.

(b) Within-sector and between-sector pairs.

Figure 3: Correlation structure in the core historical universe. Panel (a) shows the full Pearson correlation distribution. Panel (b) compares within-sector and between-sector correlations using the initial macro-sector classification.

The full distribution is centred in a moderate positive range, with mean correlation close to 0.24 and median correlation close to 0.23. This level of average dependence suggests the presence of a common component, but the distribution remains dispersed. The dispersion is important because it leaves room for sectoral and subsectoral organization instead of a single homogeneous market block.

5.4. Sectoral correlation comparison

The sector split gives a descriptive check of this interpretation. Within-sector correlations are visibly shifted to the right relative to between-sector correlations. Table 3 reports the corresponding pair counts and summary statistics. These results indicate that the initial macro-sector map is reflected in the dependency matrix.

Table 3: Within-sector and between-sector correlation summary.

Group	Pairs	Mean	Median	Std. dev.	Min	Max
Within-sector	275	0.3433	0.3082	0.1819	0.0534	0.9402
Between-sector	1378	0.2249	0.2076	0.1166	-0.0236	0.5552

The difference is economically meaningful. Within-sector pairs have mean correlation 0.3433, while between-sector pairs have mean correlation 0.2249, giving $\Delta_{\text{sector}} = 0.1184$. An asset-level label-permutation test with 10,000 sector-label reallocations preserves both the observed dependency matrix and the sector-size vector, without treating overlapping pairs as independent observations. None of the permuted statistics reaches the observed difference (null 95th percentile = 0.0245; one-sided Monte Carlo $p < 0.0001$). Conditional on exchangeability of sector labels under the null and on the initial macro-sector map, this provides evidence that the sectoral grouping is reflected in the dependency matrix.

5.5. Dynamic market synchronization

Full-sample correlations are limited because market dependence changes over time. Figure 4 shows the rolling average market correlation using 252- and 504-trading-day windows. This statistic compresses the correlation matrix into a time series that measures how synchronized the market is at each point in time. The historical panel has changing coverage in its earliest years: each window uses the assets with data available in that window, ranging from 40 assets initially to the full 58-asset universe in later periods. The series should therefore be interpreted as a descriptive measure of the average available pairwise dependence.

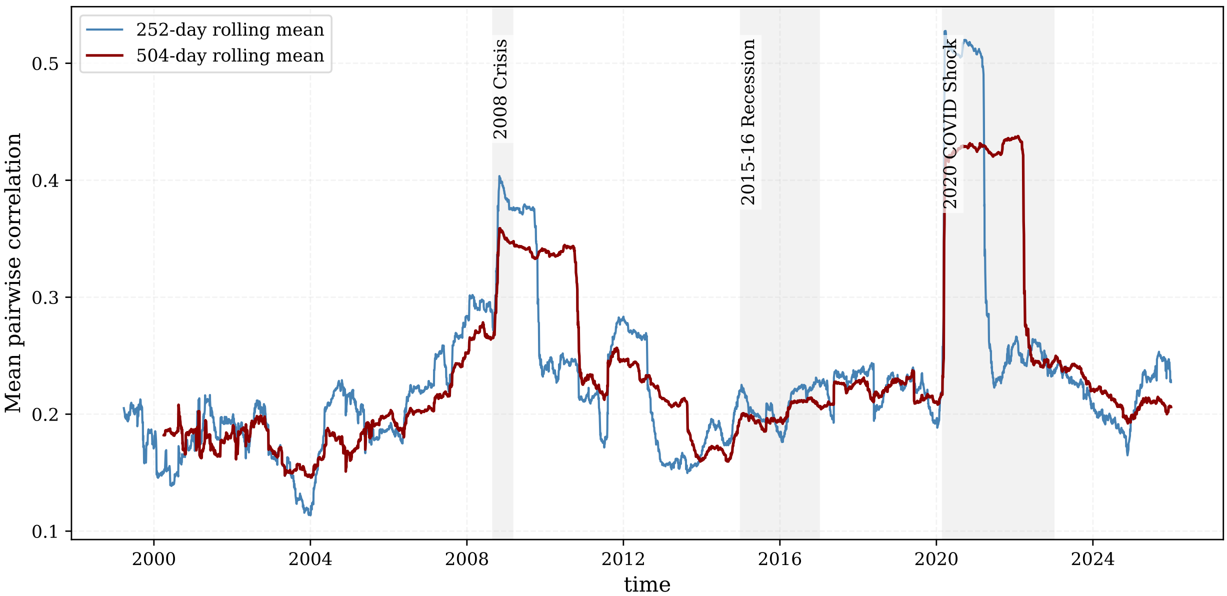


Figure 4: Rolling average market correlation for the core historical universe, estimated with 252- and 504-trading-day windows. Each window uses the assets available during that period.

The rolling average correlation tends to rise around major stress periods, including the 2008 global financial crisis, the Brazilian recession period around 2015–2016 and the 2020 COVID shock. These episodes indicate periods of stronger market-wide synchronization, which helps explain the dominant leading eigenvalue in the full-sample correlation matrix.

Figure 5 adds a more granular view by showing selected pairwise dynamic correlations under different estimators. The pairs were selected for their economic interpretability: PETR4–VALE3 compares two large commodity-exposed firms in distinct subsectors; BBDC4–BBAS3 represents financials; GGBR4–USIM5 represents steel and metallurgy; and ELET3–CMIG4 represents utilities. The figure is illustrative rather than a representative sample of all possible pairs.

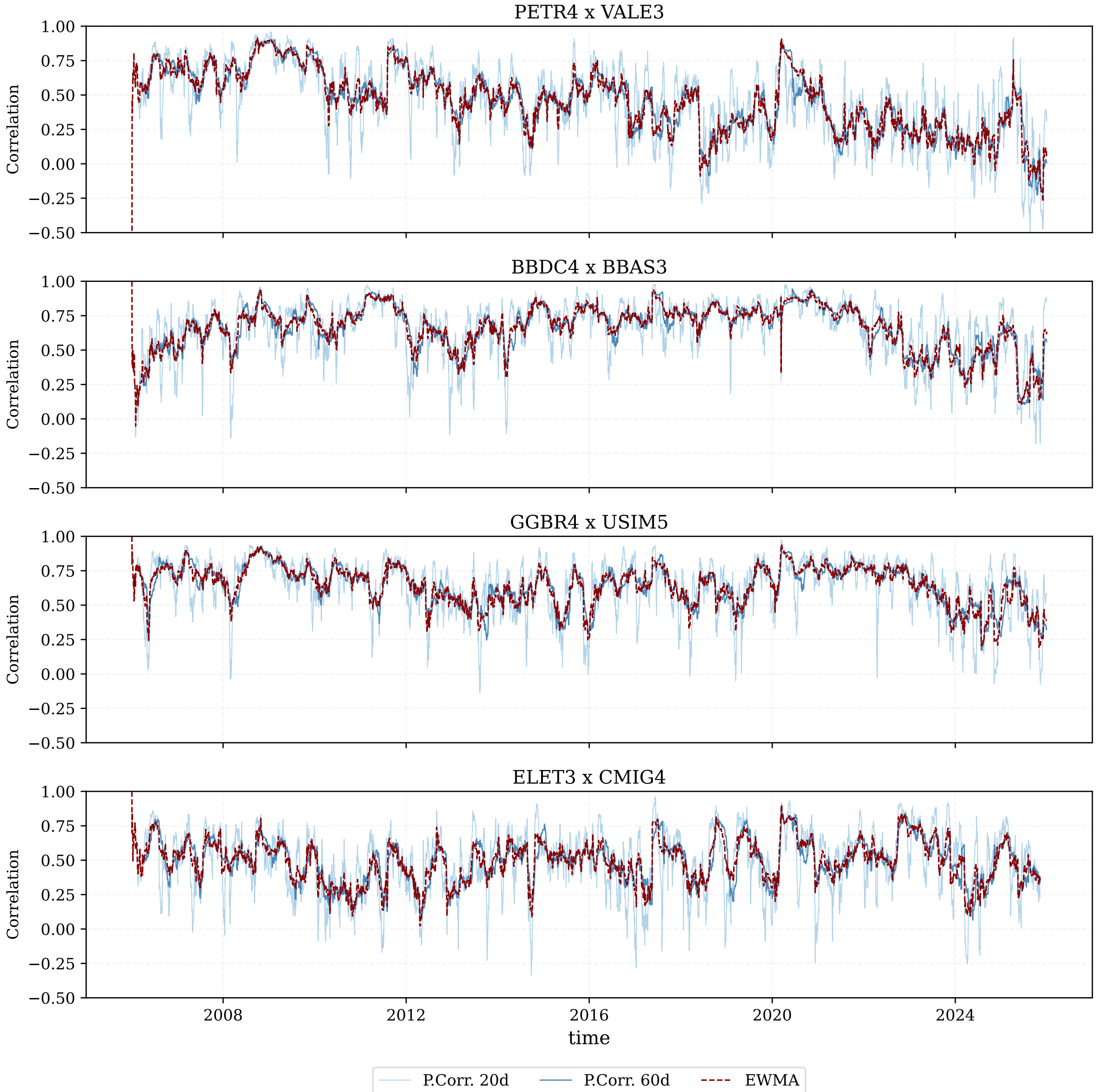


Figure 5: Dynamic pairwise correlations for selected economically interpretable asset pairs. Pearson estimates use 20- and 60-trading-day windows; EWMA uses a 20-day half-life. The estimates show how individual dependencies can vary across time scales.

The analytical role of these estimates is to show that a single full-sample correlation can conceal episodes in which a dependency strengthens, weakens or temporarily breaks down. The 20-day Pearson estimate reacts quickly to local changes but is noisy; the 60-day estimate is smoother but slower; and EWMA with a 20-day half-life provides an intermediate adaptive estimate. Together, the three views distinguish persistent co-movement from short-lived correlation bursts. This makes the figure a useful sensitivity check for the static correlation matrix used in the subsequent RMT and network analyses: it does not replace the full matrix, but demonstrates why its resulting structures should be interpreted as long-run summaries of a time-varying dependency system.

6. Random Matrix filtering

6.1. Empirical eigenspectrum

Random Matrix Theory supplies the spectral benchmark for distinguishing structured collective modes from sampling noise in the empirical correlation matrix [13, 14]. The core historical matrix uses $N = 58$ assets and $T = 1527$ observations, giving $Q = T/N \approx 26.33$. The Marčenko–Pastur bounds are $\lambda_- = 0.6482$ and $\lambda_+ = 1.4278$.

Table 4: RMT summary for the core historical universe.

N	T	Q	λ_-	λ_+	λ_1	λ_2	λ_3	λ_4	λ_5	Above λ_+
58	1527	26.33	0.6482	1.4278	21.6505	3.0957	1.7839	1.6543	1.4835	5

Table 4 reports the numerical inputs for RMT filtering. With $N = 58$ assets and $T = 1527$ complete observations, the aspect ratio is $Q = 26.33$, which implies a relatively narrow Marčenko–Pastur noise band from $\lambda_- = 0.6482$ to $\lambda_+ = 1.4278$. The first eigenvalue, $\lambda_1 = 21.6505$, is therefore not a marginal deviation from the random-matrix benchmark but a dominant collective component. The next four eigenvalues, $\lambda_2 = 3.0957$, $\lambda_3 = 1.7839$, $\lambda_4 = 1.6543$ and $\lambda_5 = 1.4835$, also exceed the upper random bound. There are 5 sample eigenvalues outside the band: these are the candidate collective modes, indicating one broad market mode and four additional modes that may be associated with group, sectoral or localized dependency structure.

Figure 6 shows the spectrum against the random benchmark. The largest eigenvalue lies far above the Marčenko–Pastur upper bound. Four additional eigenvalues also exceed the upper edge, indicating a dominant market mode and a small number of group or sector modes.

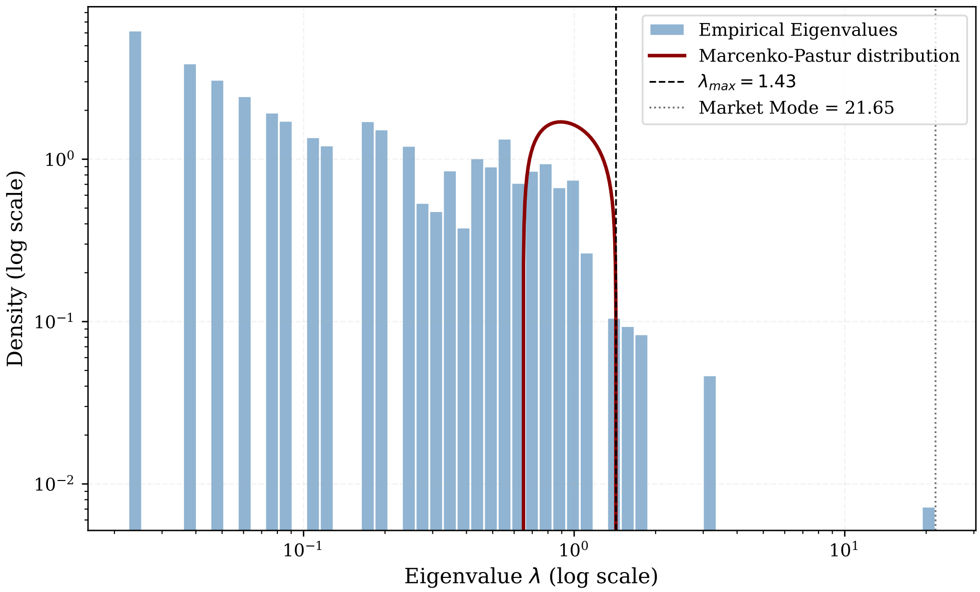


Figure 6: Empirical eigenvalue spectrum and Marčenko–Pastur bounds. The largest eigenvalue is far outside the random band and captures the market-mode component.

6.2. Market Mode and significant eigenvalues

The largest eigenvalue is 21.6505, corresponding to approximately 37.3% of the total trace of the correlation matrix. Its eigenvector has broadly positive loadings, confirming its interpretation as a market mode. It captures broad co-movement and explains why the original correlation matrix has a strong positive common component. The next four eigenvalues are 3.0957, 1.7839, 1.6543 and 1.4835. These values are above λ_+ and act as candidates for group or sector structure.

The market mode is meaningful but visually dominant. If a network is built directly from the original matrix, many edges can reflect market-wide co-movement instead of local structure. We construct the group-mode matrix to ask what remains after the broad common component is removed.

6.3. Matrix reconstruction

The reconstruction step checks that the spectral decomposition and the filtered matrices are being computed accurately. If all eigenvalue–eigenvector modes are recombined, the original correlation matrix is recovered up to floating-point precision. Table 5 reports this numerical check: the Frobenius reconstruction error is approximately 2.74×10^{-14}, and the maximum absolute reconstruction error is approximately 3.99×10^{-15}. The reported unit diagonal refers to the filtered matrix after the partial reconstruction has been rescaled for downstream correlation-based distances. This check reduces the possibility that subsequent network differences are driven by numerical reconstruction error rather than by the intended selection of market, group and noise-compatible modes.

Table 5: RMT reconstruction diagnostics.

Diagnostic	Value
Frobenius reconstruction error	2.74×10^{-14}
Maximum absolute reconstruction error	3.99×10^{-15}
C_{filtered} diagonal min	1.0000
C_{filtered} diagonal max	1.0000
Eigenvalues above λ_+	5
Market-mode trace share	0.3733

Figure 7 shows the resulting matrix components. The original-correlation panel is the complete empirical correlation matrix used throughout the paper. The market-mode matrix captures broad positive co-movement. The group-mode matrix displays weaker localized patterns after the market mode is removed. The noise component contains the random-band contribution, and the filtered matrix recombines the retained modes.

Empirically, the eigenvectors beyond the leading component show loading patterns suggesting localized or sector-related dependencies. This residual organization motivates the clustering and network sections: if localized dependency patterns remain after market-mode removal, MST and PMFG structures should differ when constructed from the group-mode matrix.

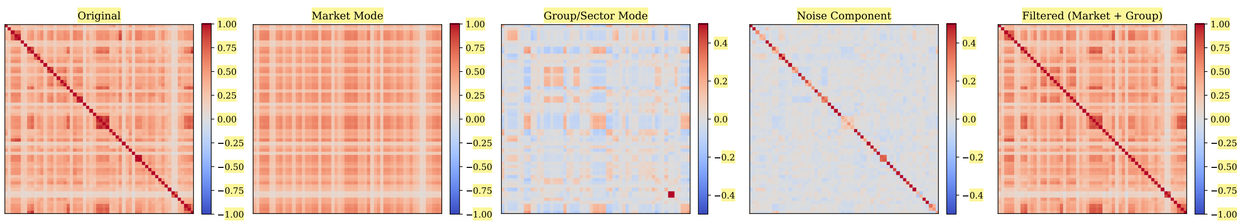


Figure 7: RMT-filtered matrix components. The original-correlation panel shows the complete 58-asset correlation matrix; the remaining panels separate it into market, group/sector, noise and filtered components.

6.4. Top eigenvector loadings

Eigenvalues identify the strength of collective modes, but eigenvectors help interpret their composition. Figure 8 reports the largest loadings for the five modes above the Marčenko–Pastur upper bound. For each eigenvector, assets are ranked by the magnitude of their loadings. The signs of eigenvector loadings are arbitrary up to multiplication by -1; the relevant information is therefore the relative contrast between groups of assets, together with the magnitude and concentration of their loadings.

The first eigenvector is a broad market mode. All 58 assets have positive loadings, ranging from 0.037 to 0.174, and the leading mode accounts for 37.3% of the trace of the correlation matrix. The largest loadings include steel, mining and financial assets, but the positive contribution is distributed across all sectors. This pattern supports its interpretation as market-wide co-movement rather than a sector-specific factor.

The second eigenvector ($\lambda_2 = 3.10$) is best read as a relative contrast between mining and steel firms—including VALE3, BRAP3/4, GGBR3/4, GOAU3/4, CSNA3 and USIM3/5—and a group led by utilities and selected consumer-cyclical assets. It therefore contains a commodity-related dimension, but it is not a pure Basic Materials factor. The third eigenvector ($\lambda_3 = 1.78$) similarly contrasts the financial pole formed by BBDC3/4, ITSA4 and BBAS3 with a more heterogeneous industrial, chemical and consumer-cyclical pole led by ROMI3, SHUL4, BRKM3, ALPA4 and GUAR3.

The fourth eigenvector ($\lambda_4 = 1.65$) is highly concentrated in TELB3 and TELB4, the two listed TELEBRAS share classes. We interpret this result as an issuer-specific share-class component, rather than as evidence of a broad telecommunications factor. The fifth eigenvector ($\lambda_5 = 1.48$) separates a coherent utilities block—including CEMIG, COPEL, Eletrobras, CPFL Energia and ISA CTEEP—from financial and selected consumer-cyclical assets. This is a plausible utilities-related pattern, but λ_5 lies only about 3.7% above the Marčenko–Pastur upper bound. Its economic interpretation should therefore remain cautious and be treated as a candidate localized mode.

Overall, Figure 8 shows that the modes beyond the market component are structured contrasts, rather than one-sector classifications. They can mix sectors when firms share macroeconomic, liquidity or balance-sheet exposures. This interpretation motivates the subsequent clustering and network analyses, which test whether these spectral contrasts reappear in other representations of the dependency matrix.

6.5. Issuer-level share-class robustness

The 58-asset panel contains 12 companies represented by two listed share classes. This can amplify issuer-specific co-movement and, in the case of TELEBRAS, produces a component that is not an economically broad sector factor. We therefore repeat the RMT calculation after retaining one deterministic representative per company, reducing the panel to 46 issuers. The retained series is the class with the most valid daily returns in the raw historical file, with alphabetical tie-breaking; this rule is stated in Section 4.

Table 6: Issuer-level RMT robustness check. The collapsed panel retains one share class per company.

Panel	N	T	Q	λ_+	Modes above λ_+	λ_1/N
Full share-class panel	58	1527	26.33	1.4278	5	0.3733
Issuer-collapsed panel	46	2207	47.98	1.3096	3	0.3363

The dominant market component is robust: its trace share remains 33.6% after issuer collapse, compared with 37.3% in the full panel. Three eigenvalues remain above the issuer-panel bound, so the market mode and two additional collective components are not artifacts of duplicated share classes. In contrast, the TELEBRAS share-class component necessarily disappears, and the two weaker full-panel modes do not remain above the issuer-panel comparison. The principal market-structure result is therefore robust, whereas interpretations of marginal localized modes should account for issuer representation.

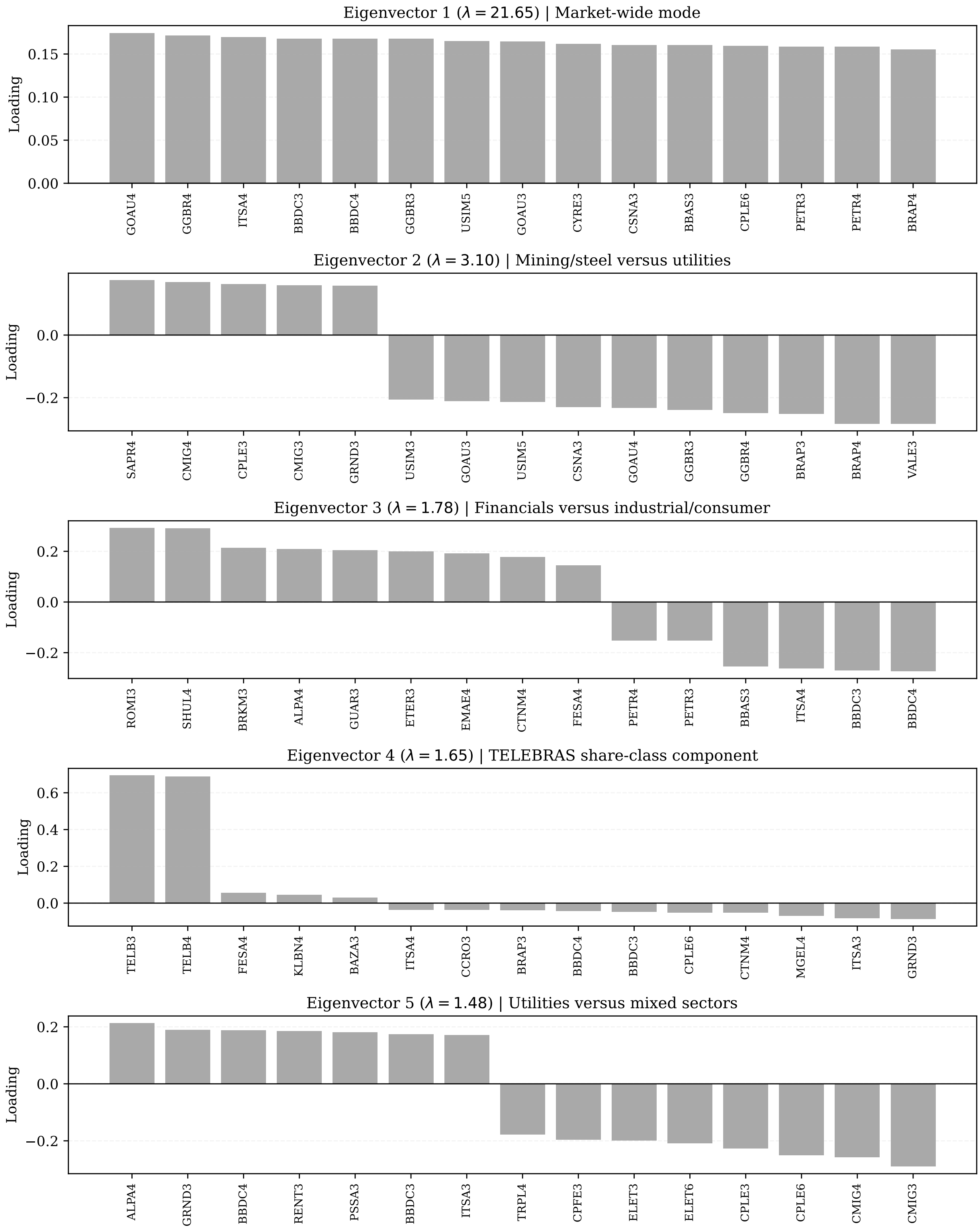


Figure 8: Top eigenvector loadings for the five RMT modes above the Marčenko–Pastur upper bound. Eigenvector 1 is a broad market mode. Subsequent eigenvectors show relative contrasts between groups of assets; their overall signs are arbitrary.

6.6. Circular-shift null robustness

The Marčenko–Pastur benchmark assumes an idealized independent-noise setting. As a complementary check, we compare the five leading eigenvalues with 1,000 independent circular-shift surrogates. Each surrogate preserves the temporal ordering and marginal distribution of every individual return series while breaking their same-date alignment. Table 7 reports rank-specific 95th-percentile thresholds and Monte Carlo exceedance probabilities.

Table 7: Circular-shift null check for the five leading eigenvalues. The null uses 1,000 independent circular shifts, one per asset and replicate.

Rank	Empirical λ	Null 95th percentile	Monte Carlo p
1	21.6505	1.6882	< 0.001
2	3.0957	1.4932	< 0.001
3	1.7839	1.3889	< 0.001
4	1.6543	1.3533	< 0.001
5	1.4835	1.3295	< 0.001

All five empirical eigenvalues exceed the rank-matched 95th-percentile circular-shift threshold, including the fifth mode. Thus, its departure from the noise bulk is not explained solely by the temporal dependence or non-Gaussian marginal behavior retained by this surrogate null. This result should be read jointly with the issuer-collapsed check: the fifth mode is statistically distinguishable from this synchronization-free null, but its detailed economic interpretation remains sensitive to how multiple share classes are represented.

7. Hierarchical clustering and ordered heatmaps

7.1. Ordered heatmaps

RMT decomposes the correlation matrix into spectral components, while ordered heatmaps offer a visual check of whether these components display block-like organization. We order the assets through hierarchical clustering in Figure 9, which displays original, filtered and group-mode structures. The heatmaps serve as diagnostic tools, assessing whether spectral filtering is associated with patterns that can be read in relation to sector and subsector classifications.

The original matrix displays a broad positive background, reflecting the large leading eigenvalue. The filtered matrix preserves part of this organized dependence while reducing the contribution of eigenvalues inside the random-matrix band. The group-mode matrix has lower average correlation by construction, but its ordered representation displays localized block-like patterns that are less dominated by the market-wide component.

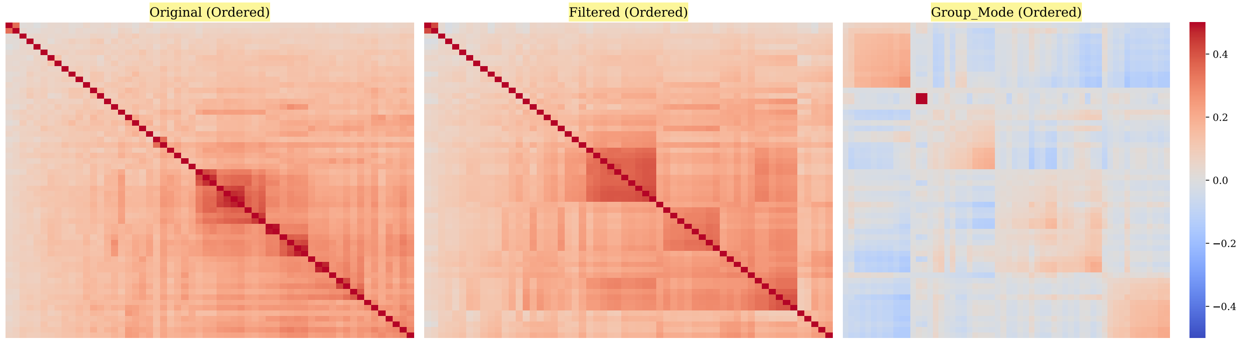


Figure 9: Ordered heatmaps for original and RMT-filtered correlation matrices. Hierarchical ordering reveals block structures that are less visible in unordered matrices.

7.2. Dendrograms comparison

Dendrograms give another view of the same distance geometry. Figure 10 compares hierarchical clustering trees for original, filtered and group-mode matrices, and Table 8 reports the corresponding cophenetic correlations. These trees help identify which assets merge at shorter distances and how the hierarchy changes after market-mode removal. We do not formally test branch stability here; the dendrograms are used as exploratory summaries of the distance geometry.

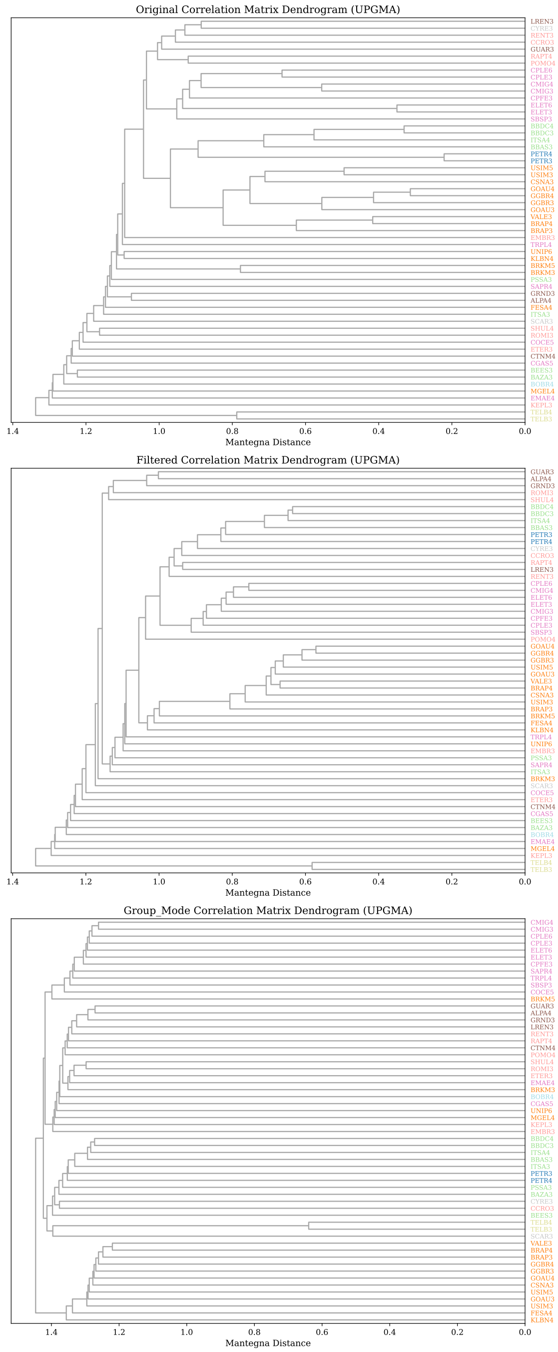


Figure 10: Dendrogram comparison for original, filtered and group-mode matrices using Mantegna distances and average linkage.

Table 8: Cophenetic correlations for hierarchical clustering.

Matrix	Assets	Cophenetic correlation
Original	58	0.9500
Filtered	58	0.9340
Group Mode	58	0.8705

The cophenetic correlation is highest for the original matrix, 0.9500, and remains high for the filtered matrix, 0.9340. The group-mode value is lower, 0.8705. This decline is expected because removing the leading market component reduces the smooth common structure that is easier to summarize hierarchically. The lower value indicates that the group-mode distance geometry is less tree-like, not that it is uninformative.

The clustering results connect naturally to network filtering. Ordered heatmaps and dendrograms suggest that localized structure remains after RMT filtering. We next examine how this structure appears when only the strongest distance-ranked edges are retained under MST and PMFG constraints.

8. Financial network topology

Clustering summarises hierarchical organisation, but it does not identify graph-theoretic hubs, bridges or local cycles. Financial networks give these additional views by retaining selected high-correlation, short-distance edges under structural constraints. We move from dendrograms to filtered financial networks to examine how dependency structure changes after RMT filtering.

8.1. MST backbone

The MST is the sparsest connected representation of the dependency geometry. With $N = 58$, it contains 57 edges. Figure 11 compares the MST built from the original matrix with the MST built from the group-mode matrix. This comparison illustrates how the MST backbone changes after the leading market component is removed.

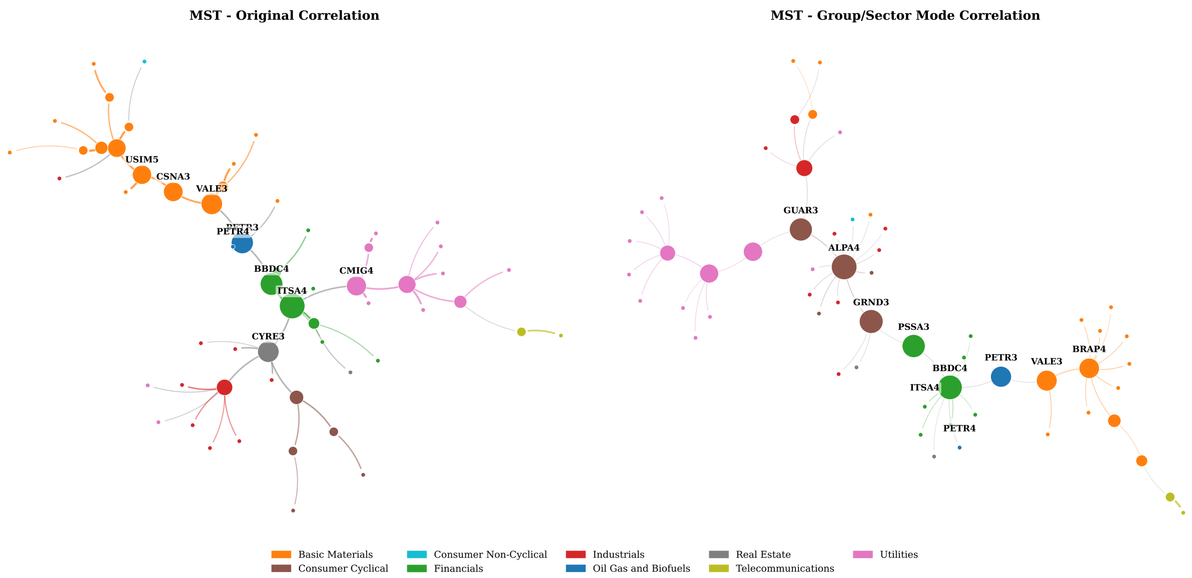


Figure 11: Refined MST comparison for original and group-mode correlation matrices. The original MST emphasizes strong market-mode dependencies, while the group-mode MST highlights weaker localized dependency channels.

The original MST has mean edge correlation 0.5796, while the group-mode MST has mean edge correlation 0.1388. This decline is expected because the group-mode network is constructed after excluding the dominant common component. The same-sector edge ratio also changes, from 0.7193 in the original MST to 0.6316 in the group-mode MST. The group-mode MST is not simply a weaker version of the original tree. It represents a different dependency backbone constructed after removing the leading common component.

8.2. PMFG topology

The PMFG is less sparse than the MST because it preserves planarity while retaining $3N - 6 = 168$ edges. It can retain cycles, triangles and 4-cliques, which are absent from the MST. Figure 12 compares original and group-mode PMFGs. This is the main network visualization in the paper because the PMFG retains enough structure to study local connectivity patterns, clustering and hub reorganization.

The original PMFG has mean edge correlation 0.5177. The group-mode PMFG has mean edge correlation 0.1109. Despite this lower average edge strength, the group-mode PMFG has higher average clustering, 0.6653, than the original PMFG, 0.5273. Removing the market mode lowers the absolute level of edge correlations, but the resulting PMFG is associated with stronger local clustering.

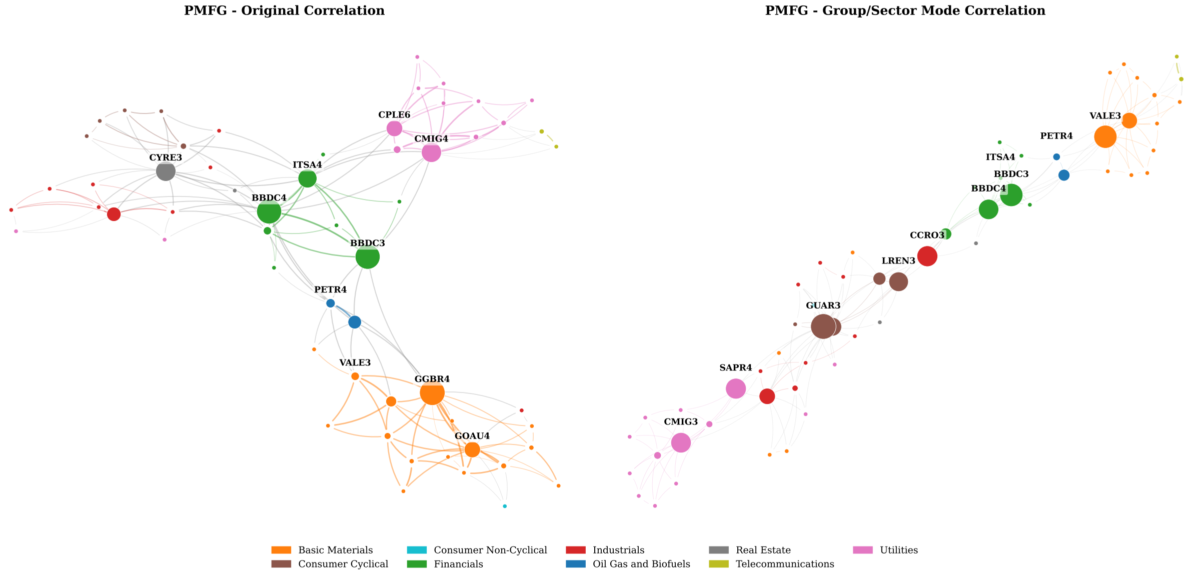


Figure 12: Refined PMFG comparison for original and group-mode correlation matrices. The group-mode PMFG reduces market-wide dominance and highlights more localized dependency channels.

8.3. Original vs Group/Sector Mode networks

Table 9 summarizes the main MST and PMFG statistics. Across both graph families, original networks display stronger average edge correlations, while group-mode networks display weaker edge correlations but retain nontrivial sectoral and graph structure. Original networks have stronger average edge correlations, as expected, because they are constructed from matrices that include the leading market component. Group-mode networks have weaker edges but retain patterns aligned with sectoral and subsectoral groupings.

Table 9: Selected MST and PMFG topology statistics.

Network	Matrix	Edges	Mean edge corr.	Same-sector ratio	Clustering	Top degree	Top betweenness	4-cliques
MST	Original	57	0.5796	0.7193	0.0000	RAPT4	ITSA4	0
MST	Group mode	57	0.1388	0.6316	0.0000	ALPA4	ALPA4	0
PMFG	Original	168	0.5177	0.6012	0.5273	BBDC4	GGBR4	55
PMFG	Group mode	168	0.1109	0.5417	0.6653	GUAR3	GUAR3	55

8.4. Temporal stability of network features

The static graphs summarize the full synchronized sample, but their components need not be stable over time. We therefore re-estimate each network in three consecutive 509-trading-day windows: 23 August 2006–13 February 2012, 28 February 2012–2 June 2021, and 4 June 2021–15 August 2024. Table 10 reports the mean pairwise overlap across the three window pairs. Edge persistence is modest in the original networks (mean Jaccard 0.206 for MST and 0.212 for PMFG) and lower in the group-mode networks (0.047 and 0.140, respectively). Top-five betweenness-hub overlap is only 0.20–0.27 on average, and community ARI values are similarly low.

No top-five betweenness hub appears in all three windows; the most recurrent hubs occur in two. Thus the full-sample graph identifies long-run average dependency channels, while its individual edges, central nodes and modular partitions are sensitive to the observation window. The lower persistence of group-mode edges is consistent with a weaker component reconstructed after removal of the dominant market mode. This result strengthens the descriptive interpretation of Figures 11–14: they reveal economically interpretable configurations in the full sample, not invariant network structure.

Table 10: Temporal stability across three consecutive 509-trading-day windows. Values are means over the three window pairs.

Network	Matrix	Edge Jaccard	Top-5 hub overlap	Community ARI
MST	Original	0.206	0.267	0.138
MST	Group mode	0.047	0.200	0.182
PMFG	Original	0.212	0.200	0.214
PMFG	Group mode	0.140	0.200	0.174

8.5. Network topology metrics

Figure 13 expands the comparison beyond average edge correlation. It summarizes mean edge correlation, same-sector ratio, same-subsector ratio and average clustering across network and matrix types. This figure illustrates that network structure cannot be reduced to a single edge-weight statistic.

The network comparison supports a subsector-level interpretation. Group-mode networks have lower mean edge correlations, while the PMFG clustering coefficient increases. Same-sector and same-subsector ratios remain substantial after market-mode removal. Table 11 confirms that all PMFG variants satisfy the expected planar clique counts. This pattern suggests localized dependency structure, although a formal comparison with graph null models would be required to quantify how unlikely the structure is under randomized alternatives. MST clustering is zero by construction because trees do not contain triangles; the PMFG clustering coefficient is the relevant statistic for local triangular organization.

Table 11: PMFG clique and planarity diagnostics.

Matrix	Nodes	Edges	Triangles	4-cliques	Planar
Original	58	168	166	55	Yes
Filtered	58	168	166	55	Yes
Group Mode	58	168	166	55	Yes

The clique counts are construction checks. For a PMFG with 58 nodes, 168 edges are expected. The presence of triangles and 4-cliques shows that the PMFG retains local graph structures that are absent from the MST. These structures can be transformed into network predictors for the forecasting experiment, although their economic interpretation depends on the underlying edge weights and sector composition.

8.6. Hub reorganization after RMT filtering

Hub rankings summarize which assets occupy central positions in the constructed networks. Figure 14 compares hub rankings before and after filtering. In the original PMFG, GGBR4 is the top betweenness node and BBDC4 is the top degree node. In the group-mode PMFG, GUAR3 becomes the leading node by both degree and betweenness.

This change in hub ranking is meaningful for interpretation. Original hubs can reflect market-wide correlation strength and broad exposure to the common component. Group-mode hubs act as localized bridges in the network obtained after removing the leading common component. The appearance of assets such as GUAR3, BBDC3, VALE3, CCRO3, SAPR4 and CMIG3 in group-mode rankings suggests that centrality depends on the matrix used to construct the network. Because hub identities can be sensitive to small changes in edge ranking, we report these rankings as descriptive summaries rather than stable structural labels.

9. Aggregated subsector dependency network

Asset-level networks can be difficult to interpret when many share classes and sectors are present. The aggregated subsector dependency network converts ticker-level dependencies into a subsector-level map. Nodes represent subsectors and edges summarize average retained dependency strength between subsectors. This is analogous to sector or country aggregation maps used in the financial dependency literature, but adapted to the Brazilian equity universe [3].

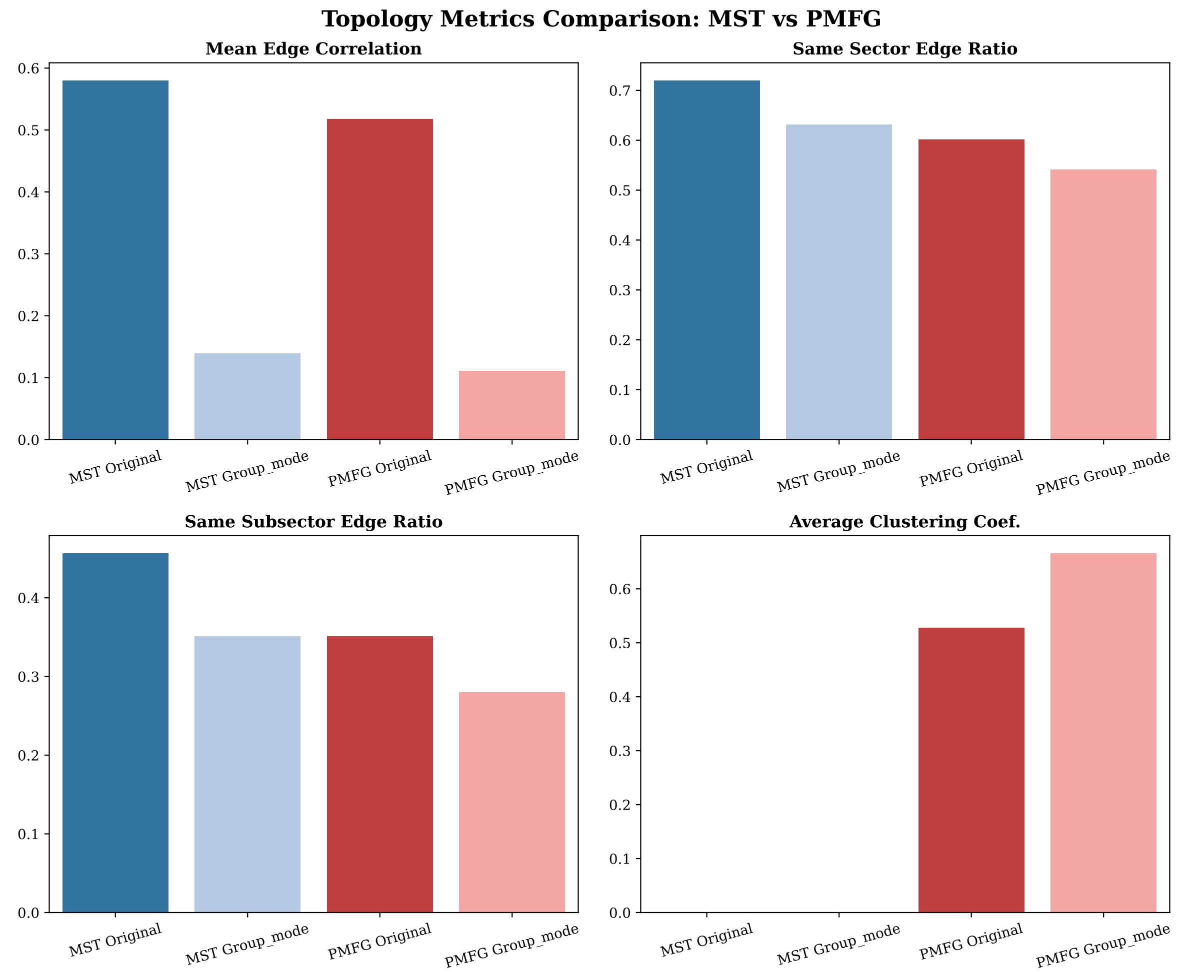


Figure 13: Network structure comparison across original, filtered and group-mode MST/PMFG representations.

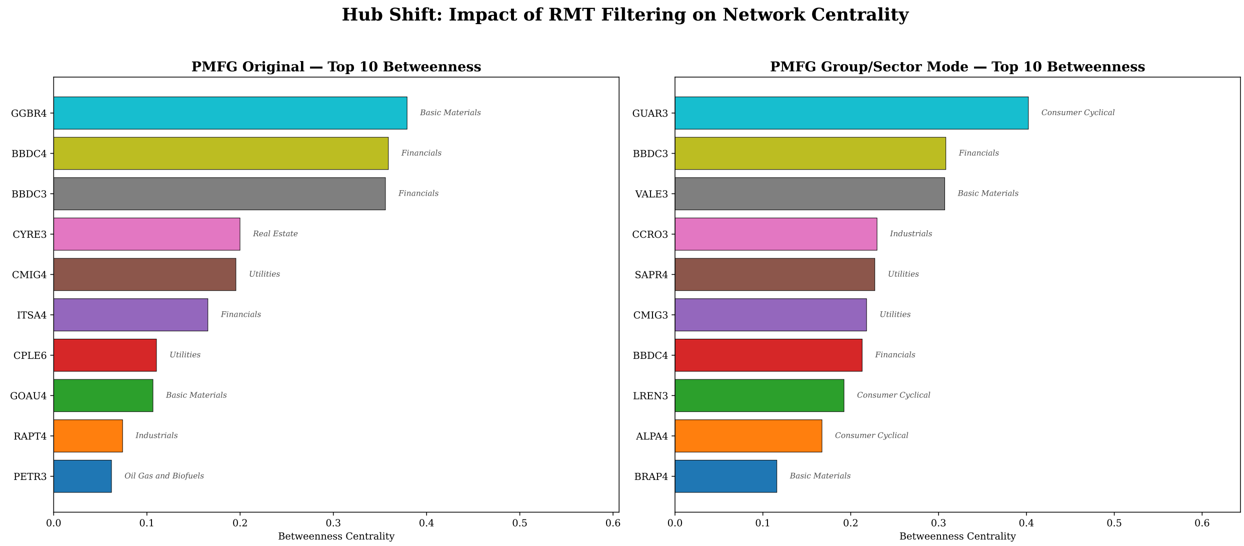


Figure 14: Hub-rank comparison across original and group-mode networks. RMT filtering changes which assets occupy central positions in the dependency network.

We summarize the resulting network in Figure 15 and Table 12. The network contains 22 subsectors and 66 edges, with density 0.2857. The mean edge dependency is 0.0566 and the median edge dependency is 0.0561. These values are lower than typical raw asset-level correlations, which is expected because the network summarizes filtered and aggregated dependencies instead of direct pairwise asset correlations.

The top-degree subsector is Retail, indicating broad direct connectivity in the aggregated dependency map. Electric Utility has the highest betweenness, suggesting that it occupies a bridge-like position in the subsector-level network. Apparel and Footwear has the highest weighted degree, reflecting stronger dependency intensity across its retained connections. These summaries help translate the ticker-level PMFG structure into economic language.

The subsector network also connects the structural results to volatility modelling. If network structure contains information beyond univariate volatility persistence, then node-level or group-level centrality measures may help explain future volatility regimes. We examine this idea in an exploratory retrospective experiment; a real-time test requires the network to be reconstructed at each forecast origin.

Table 12: Aggregated subsector network summary.

Subsectors	Edges	Density	Mean edge dep.	Median edge dep.	Top degree	Top betweenness	Top weighted degree
22	66	0.2857	0.0566	0.0561	Retail	Electric Utility	Apparel and Footwear

Notes: Edges are retained subsector-level links in the aggregated dependency network. Density is the ratio between retained and possible undirected subsector links. Mean and median edge dependency are computed from the retained edge weights W_{ab} defined in Section 4.10. Degree counts retained links, weighted degree sums retained edge weights and betweenness is the standard shortest-path betweenness centrality computed on the aggregated graph.

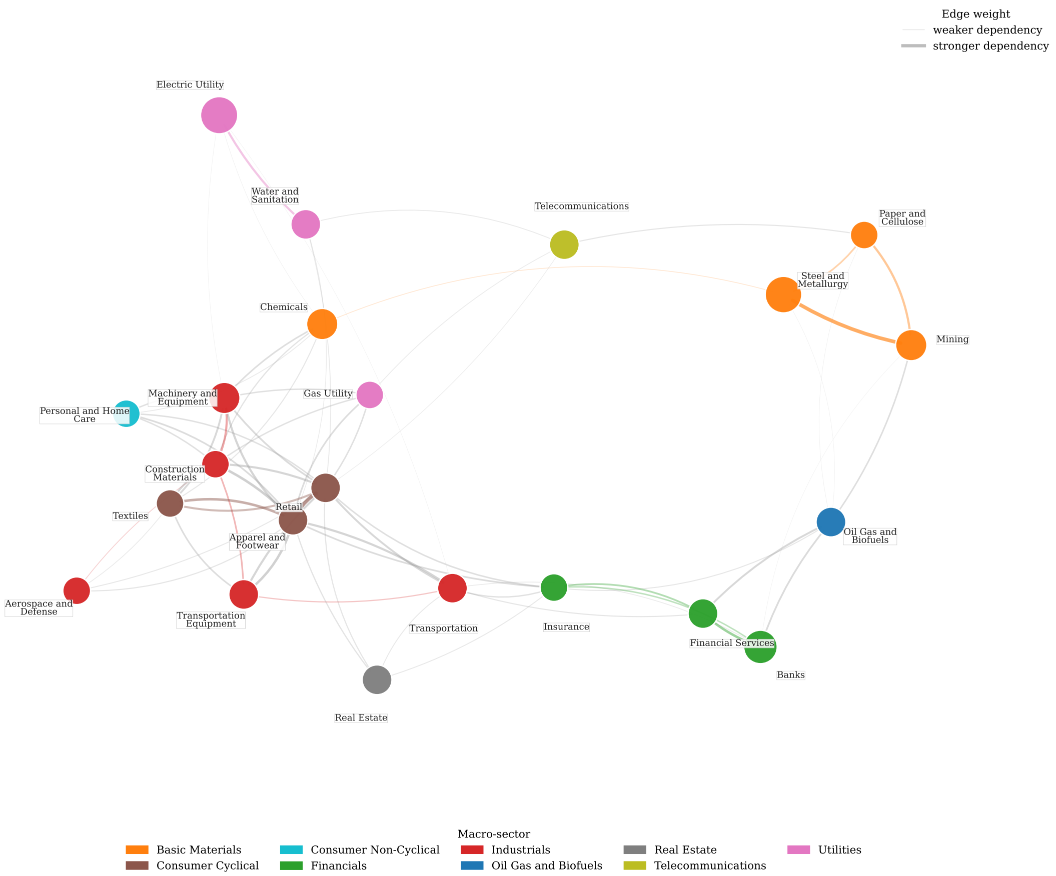


Figure 15: Aggregated subsector dependency network. Nodes represent subsectors, node colour identifies the macro-sector and node size is proportional to the number of assets in the subsector. Edges represent retained subsector-level dependency weights, with thicker edges indicating stronger average dependency.

10. Volatility forecasting

10.1. Chronological evaluation with ex-post structural descriptors

We conduct a retrospective evaluation of whether RMT and network descriptors are associated with realized-volatility model performance. The experiment uses PETR4, VALE3 and BBDC4 as target assets and compares machine-learning models with simple neural extensions. The data are split chronologically: training covers 2006–2017, validation covers 2018–2020 and testing covers 2021–2025. This ordering keeps model fitting and feature-set selection separated from the held-out period. However, the RMT and network descriptors are calculated from the full synchronized panel and broadcast across dates; Sets B and C consequently use ex-post structural information. Their test-period scores are not evidence of a tradable, real-time forecasting advantage. They are an exploratory added-information comparison that a recursive rolling design must validate.

10.2. Machine-learning and deep-learning models

The machine-learning models use three nested feature sets in an ablation design. Set A contains standard lagged return and volatility features. Set B adds rolling market variables and full-sample RMT descriptors. Set C adds full-sample MST and PMFG descriptors. This nesting allows the experiment to assess whether ex-post network structure is associated with model performance beyond lagged-volatility persistence and broad market/RMT descriptors.

The deep-learning extension uses two simple architectures. The MLP is a tabular neural regressor trained on the same feature sets as the machine-learning models. The CNN-1D uses short rolling windows of return and volatility variables. Both neural models use positive-output regression for realized volatility. These models are included as an incremental extension, not as a claim that the neural architecture has been optimized for this market.

10.3. Chronological test-period model comparison

Figure 16 summarizes held-out test-period QLIKE loss with moving-block bootstrap intervals, and Table 13 reports the aggregated test-period metrics. Lower QLIKE is better. For each model and horizon, the reported feature set is selected using the validation period and then evaluated on the test period. The temporal holdout applies to model fitting and selection; it does not make the full-sample structural descriptors point-in-time features.

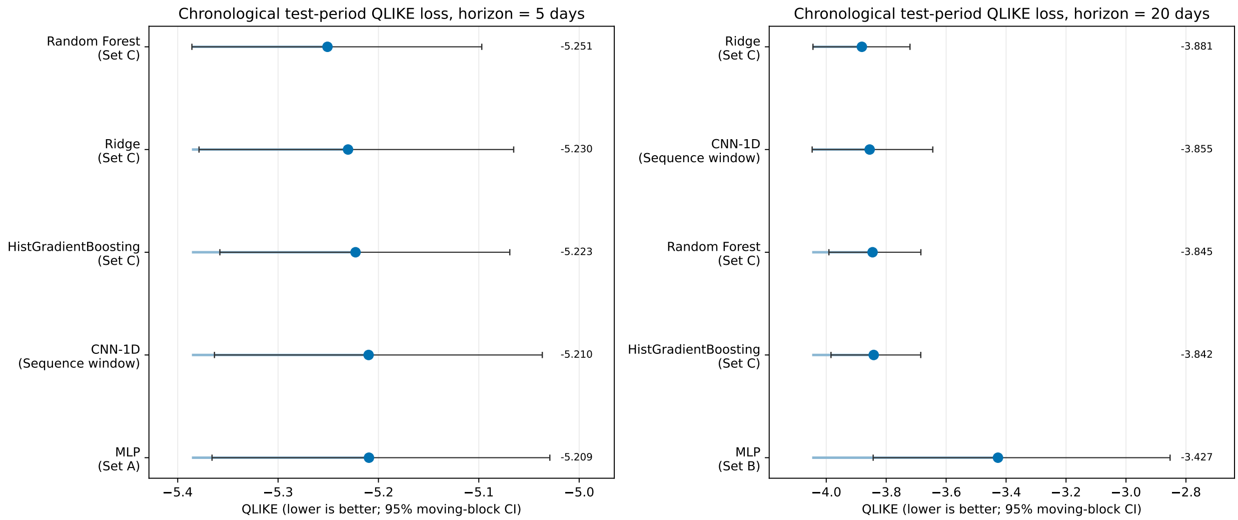


Figure 16: Chronological test-period comparison for machine-learning and deep-learning volatility models. Points are mean QLIKE across PETR4, VALE3 and BBDC4; horizontal bars are 95% circular moving-block bootstrap intervals with block length equal to the horizon. Labels show the validation-selected feature set for each model and horizon. Sets B and C include ex-post full-sample structural descriptors and are not real-time forecasting specifications.

At the 5-day horizon, Random Forest with Set C obtains the lowest average test-period QLIKE among the validation-selected models. Ridge, CNN-1D and MLP remain close in MAE and RMSE, and the 95% moving-block QLIKE intervals overlap materially across the leading specifications. Thus the ranking is descriptive rather than evidence of a precisely estimated performance separation. Per-asset metrics and intervals are reported in the supplementary material.

At the 20-day horizon, Ridge with Set C obtains the lowest average test-period QLIKE, while CNN-1D gives competitive MAE, RMSE and held-out R^2. The MLP has relatively low MAE and RMSE but weaker QLIKE, indicating that it can track average volatility levels while still producing less favourable variance-proxy loss. The focused HAC loss-difference check in Table 14 compares the validation-selected structural champion (Ridge Set C at both horizons) with the validation-selected point-in-time Set A baseline (Ridge). The mean baseline-minus-structural difference is negative at both horizons, and the one-sided tests do not support lower structural QLIKE. Accordingly, the retrospective exercise does not establish a forecasting gain from the ex-post descriptors; a recursive feature construction remains necessary for that claim.

Table 13: Aggregated chronological test-period comparison (2021–2025). Values are averaged across PETR4, VALE3 and BBDC4 after validation-based model selection. Sets B and C use ex-post full-sample structural descriptors.

Horizon	Model	Feature set	MAE	RMSE	QLIKE	95% CI	R^2_{oos}
5	Random Forest	Set C (+Network)	0.0161	0.0227	−5.2508	[−5.3859, −5.0970]	0.1989
5	Ridge	Set C (+Network)	0.0146	0.0214	−5.2303	[−5.3787, −5.0652]	0.2774
5	HistGradientBoosting	Set C (+Network)	0.0168	0.0234	−5.2228	[−5.3580, −5.0691]	0.1477
5	CNN-1D	Sequence window	0.0143	0.0213	−5.2098	[−5.3634, −5.0367]	0.2886
5	MLP	Set A (Classical)	0.0138	0.0210	−5.2095	[−5.3658, −5.0292]	0.3046
20	Ridge	Set C (+Network)	0.0219	0.0309	−3.8812	[−4.0443, −3.7206]	0.3803
20	CNN-1D	Sequence window	0.0205	0.0306	−3.8552	[−4.0470, −3.6445]	0.3951
20	Random Forest	Set C (+Network)	0.0264	0.0368	−3.8455	[−3.9911, −3.6843]	0.1667
20	HistGradientBoosting	Set C (+Network)	0.0265	0.0371	−3.8417	[−3.9841, −3.6847]	0.1685
20	MLP	Set B (+Market/RMT)	0.0213	0.0305	−3.4271	[−3.8434, −2.8528]	0.3726

Notes: MAE is mean absolute error, RMSE is root mean squared error, QLIKE is the variance-proxy loss defined in Section 4.13, and R^2_{oos} is computed relative to the training-sample mean benchmark. The QLIKE interval is a 95% circular moving-block bootstrap interval based on 2,000 replications and a block length equal to the horizon. Set A contains point-in-time return and volatility predictors. Set B adds rolling market variables and full-sample RMT descriptors; Set C adds full-sample MST/PMFG descriptors. Thus the table evaluates chronological model fitting and selection, not a strict real-time forecast implementation for Sets B and C.

Table 14: HAC loss-difference test for the validation-selected structural specification against the validation-selected Set A baseline. Positive differences favour the structural specification.

Horizon	Structural	Set A baseline	Mean difference	HAC s.e.	Statistic	One-sided p
5	Ridge Set C	Ridge Set A	−0.0045	0.0031	−1.47	0.929
20	Ridge Set C	Ridge Set A	−0.0072	0.0039	−1.86	0.969

Notes: The loss differential is the date-level mean of $\text{QLIKE}_t^A - \text{QLIKE}_t^{\text{struct}}$ across the three assets. The HAC variance uses Bartlett/Newey–West lags 4 and 19 for the 5- and 20-day horizons, respectively. The one-sided alternative is that the structural loss is lower.

10.4. Realized vs predicted volatility checks

Aggregate metrics can hide important time-series behaviour. Figure 17 compares realized 20-day volatility with the validation-selected machine-learning or deep-learning forecast for PETR4, VALE3 and BBDC4. We include this plot in the main text because it illustrates how the selected model tracks changing volatility regimes.

The plots suggest that broad volatility regimes are tracked reasonably well, but extreme spikes remain difficult to forecast. This matches the aggregate results: flexible models can be competitive, but lagged volatility remains the main source of point-in-time predictive information. The network variables should be read as ex-post structural descriptors rather than replacements for time-series volatility information.

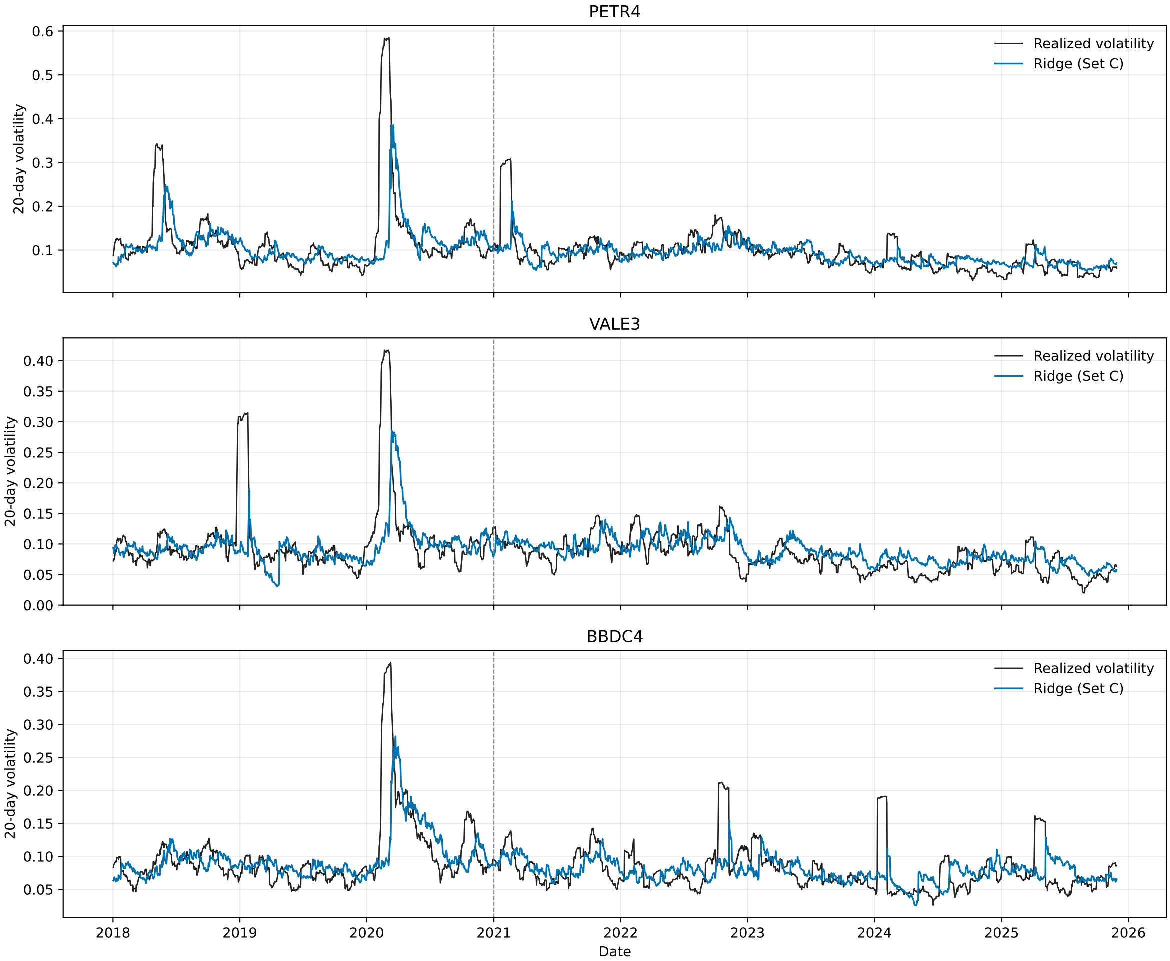


Figure 17: Realized versus predicted 20-day volatility for the validation-selected machine-learning or deep-learning model. The vertical dashed line marks the beginning of the 2021–2025 test period.

10.5. Feature importance and network signal

Random Forest feature importance gives a model-specific check of which predictors contribute most to the fitted model. Figure 18 shows that standard volatility features dominate the model. Rolling absolute returns, lagged realized volatility, rolling volatility and EWMA variables carry most of the importance.

The strongest features are rolling absolute return over 20 days, lagged realized volatility and longer rolling volatility measures. Network descriptors have smaller importance values than standard volatility predictors, but selected PMFG variables enter the top-ranked predictors, especially at the 20-day horizon. Because Random Forest importance measures can be affected by correlation among predictors, and because the structural descriptors are ex post, these rankings are model checks rather than causal or real-time measures of variable relevance. In summary, topology does not replace lagged-volatility features; its point-in-time contribution remains to be tested with recursively reconstructed features.

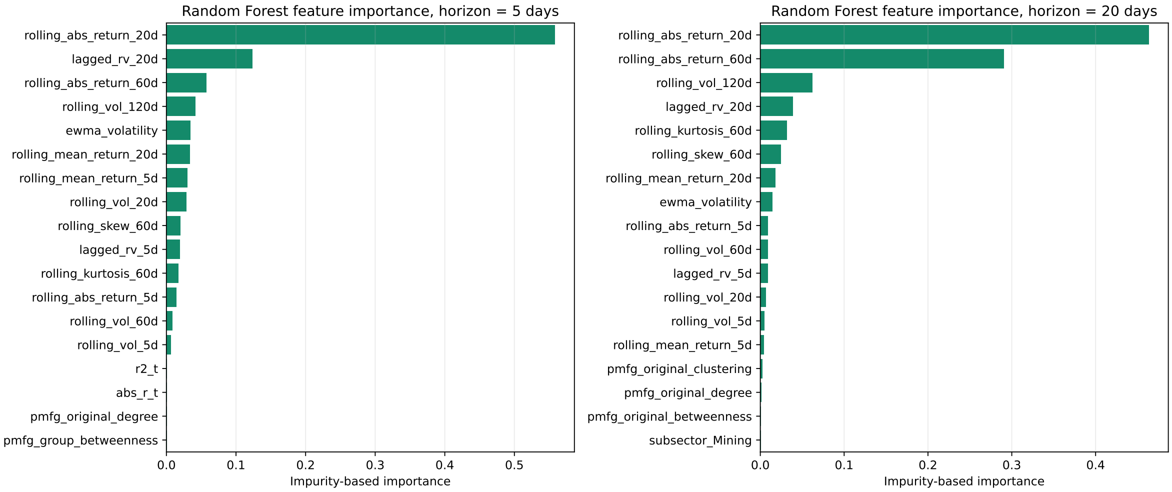


Figure 18: Feature importance for Random Forest volatility forecasting models. Standard volatility predictors dominate the selected rankings, while selected PMFG-based and sector/subsector variables appear with smaller but nonzero importance in the network-augmented feature set.

11. Discussion

Taken together, the results point to a layered interpretation of Brazilian equity dependencies, in line with the multi-scale perspective advocated by Raddant and Di Matteo [3]. The first layer is a market-mode component. It is reflected in the moderate positive average correlation ($\bar{\rho} \approx 0.24$), the increase in rolling average correlation around stress periods and the very large leading eigenvalue ($\lambda_1 = 21.65$, approximately 37.3% of the trace of the correlation matrix). We interpret this component as broad market synchronization. In the Brazilian context, such synchronization may reflect common exposure to domestic monetary conditions, exchange-rate dynamics, country risk and global commodity cycles, although these channels are not separately identified in the present analysis. The component is also methodologically important because it can dominate raw networks and obscure weaker local structures.

The second layer consists of candidate group and sector structure. Within-sector correlations are higher than between-sector correlations on average, and the RMT decomposition identifies four additional eigenvalues above the random upper edge. Eigenvector loadings and group-mode matrices suggest that commodity, financial, telecommunications and utility-related structures contribute to the non-random part of the spectrum. This reading is descriptive, not causal: we identify structure in the dependency matrix, not the economic mechanisms that generate each edge.

The third layer is noise-compatible residual dependence. Not every correlation is economically meaningful, and not every edge retained by a graph is a stable economic relationship. This is why we combine RMT, heatmaps, dendrograms and network metrics. A result is more credible when it appears across several representations, not only in one visualization.

The network results show that market-mode removal changes structure in a systematic way. Original networks have stronger average edge correlations, while group-mode networks have much lower average edge correlations. However, the group-mode PMFG has higher clustering (0.6653 versus 0.5273). Notably, lower average edge strength does not mean absence of structure. It means that the dominant common component has been removed and localized organization becomes easier to observe [17, 18]. The PMFG is well suited for this analysis because it preserves richer graph structure than the MST, retaining 168 edges with 166 triangles and 55 4-cliques that allow hub-rank shifts, clustering changes and subsector dependency maps to be studied in detail.

Clustering summarizes hierarchical organization, but it does not identify graph-theoretic hubs, bridges or local cycles. The transition from dendrograms to filtered financial networks adds information. Hub reorganization after RMT filtering is meaningful: original hubs such as GGBR4 and BBDC4 are central in networks that include the common component, while group-mode hubs such as GUAR3 occupy central positions after the leading component is removed. This suggests that centrality in financial networks depends on the matrix used to construct the network.

The volatility-modelling results require caution. Machine-learning and simple deep-learning models do not produce uniform differences across horizons and metrics. Instead, models augmented with ex-post network descriptors are competitive at selected horizons, while lagged volatility variables remain dominant. Because the RMT and network descriptors are estimated from the full sample, these results identify a retrospective structural association rather than a real-time forecast gain. The net interpretation is modest: topology summarizes cross-sectional geometry that may complement the temporal dynamics of individual asset variance, but its point-in-time value remains untested.

The broader contribution is methodological. We connect econophysics filtering, financial networks and volatility modelling in a single reproducible workflow for B3 equities. By documenting each step transparently and separating chronological model fitting from ex-post structural enrichment, the workflow offers a template for a stricter next step: rolling or expanding-window RMT and network reconstruction at each forecast origin. That extension can be applied to other emerging markets, different time scales and more sophisticated forecasting architectures.

12. Limitations

The first limitation concerns time variation and information availability. The RMT and network features used in the initial volatility-modelling exercise are estimated from the full synchronized sample and then broadcast across dates. This construction uses information unavailable at an historical forecast origin; consequently, the results for feature sets containing these descriptors cannot be interpreted as strict out-of-sample or real-time forecast performance. They are useful for identifying an ex-post structural association, but a valid forecasting test must reconstruct RMT and network features recursively, using only observations available at each origin, and re-select models within an expanding or rolling evaluation design.

The second limitation concerns the target set. We use PETR4, VALE3 and BBDC4 in the forecasting experiment. These assets are liquid and economically important, but they do not cover the full B3 cross-section. Future work should extend the forecasting exercise to a larger set of assets and evaluate whether network features help more for some sectors than others.

The third limitation concerns data quality. The core historical universe is sufficiently clean for the current analysis, but broader or more liquid modern universes may contain extreme adjusted-return events that require additional validation. These events may reflect corporate actions, ticker transitions, units or adjustment-factor issues. We would flag, inspect and either correct or handle them through robustness checks before using broader universes for final network claims.

The fourth limitation concerns sector classification and issuer representation. The macro-sector and subsector map is an initial classification. It is adequate for testing whether within-sector correlations exceed between-sector correlations, but it should be reviewed ticker by ticker before submission. We provide an issuer-collapsed RMT robustness check that removes duplicate share classes; nevertheless, the deterministic representative rule is not a substitute for a fully specified liquidity, free-float or economic-exposure rule. This remains especially important for holding companies, units and firms with mixed exposure across sectors.

Finally, the forecasting analysis is limited to three target assets, static network features and simple neural architectures. Feature importance in tree models is useful but does not establish causal mechanisms. PMFG features show nonzero but modest importance, and the magnitude is small compared with lagged-volatility predictors. Future work should include rolling network features, a broader target asset set, richer temporal neural architectures and graph neural networks.

13. Conclusion

We examined the dependency structure of Brazilian equities traded on B3 using an econophysics workflow. The raw historical universe covers 1998–2025; the RMT, clustering and network findings are based on its complete 58-asset panel from 23 August 2006 to 15 August 2024. The main findings can be summarized along four dimensions.

First, the stylized-fact and correlation analysis shows that Brazilian equities display signatures commonly associated with complex financial systems: heavy-tailed return behaviour, volatility clustering, persistent absolute-return dependence and a moderate but positive average cross-correlation ($\bar{\rho} \approx 0.24$). Within-sector correlations are larger than between-sector correlations on average, confirming that sectoral organization is reflected in the dependency matrix.

Second, Random Matrix Theory decomposes the correlation matrix. Five eigenvalues exceed the Marčenko–Pastur upper bound, with the largest eigenvalue accounting for approximately 37.3% of the trace of the correlation matrix. The leading component captures a market-mode component, while the remaining eigenvectors outside the random band suggest commodity, financial and utility-related group structures. The reconstruction checks indicate numerical accuracy within floating-point precision, and the group-mode matrix retains localized dependency patterns after the dominant market component is removed.

Third, financial network analysis through MST and PMFG representations shows that RMT filtering changes the structure of asset dependencies. The group-mode PMFG has lower mean edge correlation but higher clustering than the original PMFG, indicating stronger local clustering after market-mode removal. Hub rankings shift from networks dominated by broad market co-movement to networks where different assets occupy central positions after excluding the leading component. The PMFG retains 168 edges with triangles and 4-cliques, leaving local graph structures that are absent from the MST and useful for subsector-level analysis.

Fourth, the chronological volatility-modelling experiment identifies a limited ex-post association between network topology and model performance. Models augmented with full-sample network descriptors obtain the lowest average QLIKE in selected model–horizon comparisons, while the CNN-1D model is competitive in MAE, RMSE and held-out R^2. Lagged-volatility features remain dominant, and selected PMFG centralities have smaller importance. These comparisons do not establish a real-time forecasting benefit because the RMT and network descriptors use full-sample information.

We view the net contribution of network descriptors as methodological and incremental: they do not replace lagged-volatility dynamics, but they summarize structural information from the cross-sectional geometry of the market. The essential next step is to construct RMT and network features recursively at each forecast origin, then repeat model selection and evaluation without future information. That design, together with a broader target-asset set and richer temporal neural or graph architectures, can determine whether the structural association survives as a real-time forecasting benefit.

A. Supplementary figures

Additional figures and tables generated by the computational workflow are retained as supplementary material. These include asset-universe tables, return summaries, correlation matrices, RMT eigenvector loadings, network edge lists, clique lists, centrality rankings and detailed forecasting results by asset, horizon, model and feature set.

B. Supplementary tables

The following tables are generated from the computational outputs by the supplementary-table generation script. They provide the complete classified asset universe, the deterministic issuer/share-class selection, spectral loadings, null tests and asset-level forecast metrics used in the manuscript.
All numerical entries below are generated from the CSV files named in the captions.

B.1. Asset universe

Table 15: Asset universe

Symbol	Company	Macro-sector	Subsector
ALPA4	ALPARGATAS	Consumer Cyclical	Apparel and Footwear
BAZA3	BANCO DA AMAZONIA	Financials	Banks
BBAS3	BANCO DO BRASIL	Financials	Banks
BBDC3	BRADESCO	Financials	Banks
BBDC4	BRADESCO	Financials	Banks
BEES3	BANESTES	Financials	Banks
BOBR4	BOMBRIL	Consumer Non-Cyclical	Personal and Home Care
BRAP3	BRADESPAR	Basic Materials	Mining
BRAP4	BRADESPAR	Basic Materials	Mining
BRKM3	BRASKEM	Basic Materials	Chemicals
BRKM5	BRASKEM	Basic Materials	Chemicals
CCRO3	CCR	Industrials	Transportation
CGAS5	COMGAS	Utilities	Gas Utility
CMIG3	CEMIG	Utilities	Electric Utility
CMIG4	CEMIG	Utilities	Electric Utility

Symbol	Company	Macro-sector	Subsector
COCE5	COELCE	Utilities	Electric Utility
CPFE3	CPFL ENERGIA	Utilities	Electric Utility
CPLE3	COPEL	Utilities	Electric Utility
CPLE6	COPEL	Utilities	Electric Utility
CSNA3	CSN	Basic Materials	Steel and Metallurgy
CTNM4	COTEMINAS	Consumer Cyclical	Textiles
CYRE3	CYRELA	Real Estate	Real Estate
ELET3	ELETROBRAS	Utilities	Electric Utility
ELET6	ELETROBRAS	Utilities	Electric Utility
EMAE4	EMAE	Utilities	Electric Utility
EMBR3	EMBRAER	Industrials	Aerospace and Defense
ETER3	ETERNIT	Industrials	Construction Materials
FESA4	FERBASA	Basic Materials	Steel and Metallurgy
GGBR3	GERDAU	Basic Materials	Steel and Metallurgy
GGBR4	GERDAU	Basic Materials	Steel and Metallurgy
GOAU3	METALURGICA GERDAU	Basic Materials	Steel and Metallurgy
GOAU4	METALURGICA GERDAU	Basic Materials	Steel and Metallurgy
GRND3	GRENDENE	Consumer Cyclical	Apparel and Footwear
GUAR3	GUARARAPES	Consumer Cyclical	Retail
ITSA3	ITAUSA	Financials	Financial Services
ITSA4	ITAUSA	Financials	Financial Services
KEPL3	KEPLER WEBER	Industrials	Machinery and Equipment
KLBN4	KLABIN	Basic Materials	Paper and Cellulose
LREN3	LOJAS RENNER	Consumer Cyclical	Retail
MGEL4	MANGELS	Basic Materials	Steel and Metallurgy
PETR3	PETROBRAS	Oil Gas and Biofuels	Oil Gas and Biofuels
PETR4	PETROBRAS	Oil Gas and Biofuels	Oil Gas and Biofuels
POMO4	MARCOPOLO	Industrials	Transportation Equipment
PSSA3	PORTO SEGURO	Financials	Insurance
RAPT4	RANDON	Industrials	Transportation Equipment
RENT3	LOCALIZA	Industrials	Transportation
ROMI3	INDS ROMI	Industrials	Machinery and Equipment
SAPR4	SANEPAR	Utilities	Water and Sanitation
SBSP3	SABESP	Utilities	Water and Sanitation
SCAR3	SAO CARLOS	Real Estate	Real Estate
SHUL4	SCHULZ	Industrials	Machinery and Equipment
TELB3	TELEBRAS	Telecommunications	Telecommunications
TELB4	TELEBRAS	Telecommunications	Telecommunications
TRPL4	ISA CTEEP	Utilities	Electric Utility
UNIP6	UNIPAR	Basic Materials	Chemicals
USIM3	USIMINAS	Basic Materials	Steel and Metallurgy
USIM5	USIMINAS	Basic Materials	Steel and Metallurgy
VALE3	VALE	Basic Materials	Mining

B.2. Issuer/share-class selection

Table 16: Issuer/share-class selection

Company	Symbol	Valid returns	Retained
BRADESCO	BBDC3	6885	No
BRADESCO	BBDC4	6886	Yes
BRADESPAR	BRAP3	6210	No
BRADESPAR	BRAP4	6291	Yes
BRASKEM	BRKM3	5048	No
BRASKEM	BRKM5	5783	Yes

Company	Symbol	Valid returns	Retained
CEMIG	CMIG3	6880	No
CEMIG	CMIG4	6885	Yes
COPEL	CPLE3	6787	No
COPEL	CPLE6	6852	Yes
ELETROBRAS	ELET3	6852	Yes
ELETROBRAS	ELET6	6852	No
GERDAU	GGBR3	5682	No
GERDAU	GGBR4	6174	Yes
ITAUSA	ITSA3	5710	No
ITAUSA	ITSA4	6885	Yes
METALURGICA GERDAU	GOAU3	5852	No
METALURGICA GERDAU	GOAU4	6885	Yes
PETROBRAS	PETR3	6885	Yes
PETROBRAS	PETR4	6885	No
TELEBRAS	TELB3	5386	No
TELEBRAS	TELB4	6852	Yes
USIMINAS	USIM3	6005	No
USIMINAS	USIM5	6669	Yes

B.3. Largest absolute eigenvector loadings

Table 17: Largest absolute eigenvector loadings

Rank	Eigenvalue	Symbol	Sector	$\|u\|$
1	21.6505	GOAU4	Basic Materials	0.1743
1	21.6505	GGBR4	Basic Materials	0.1713
1	21.6505	ITSA4	Financials	0.1694
1	21.6505	BBDC3	Financials	0.1676
1	21.6505	BBDC4	Financials	0.1675
2	3.0957	VALE3	Basic Materials	0.2831
2	3.0957	BRAP4	Basic Materials	0.2831
2	3.0957	BRAP3	Basic Materials	0.2507
2	3.0957	GGBR4	Basic Materials	0.2487
2	3.0957	GGBR3	Basic Materials	0.2387
3	1.7839	ROMI3	Industrials	0.2922
3	1.7839	SHUL4	Industrials	0.2903
3	1.7839	BBDC4	Financials	0.2735
3	1.7839	BBDC3	Financials	0.2689
3	1.7839	ITSA4	Financials	0.2623
4	1.6543	TELB3	Telecommunications	0.6936
4	1.6543	TELB4	Telecommunications	0.6882
4	1.6543	GRND3	Consumer Cyclical	0.0859
4	1.6543	ITSA3	Financials	0.0808
4	1.6543	MGEL4	Basic Materials	0.0694
5	1.4835	CMIG3	Utilities	0.2901
5	1.4835	CMIG4	Utilities	0.2577
5	1.4835	CPLE6	Utilities	0.2502
5	1.4835	CPLE3	Utilities	0.2265
5	1.4835	ALPA4	Consumer Cyclical	0.2129

B.4. Circular-shift RMT null test

Table 18: Circular-shift RMT null test

Rank	Empirical λ	Null 95th pct.	Exceedances	p
1	21.6505	1.6882	0	0.0010
2	3.0957	1.4932	0	0.0010
3	1.7839	1.3889	0	0.0010
4	1.6543	1.3533	0	0.0010
5	1.4835	1.3295	0	0.0010

B.5. Asset-level sector-label permutation test

Table 19: Asset-level sector-label permutation test

Assets	Pairs	Permutations	Δ_{sector}	Null 95th pct.	p
58	1653	10000	0.1184	0.0245	0.0001

B.6. Forecast metrics by asset and horizon

Table 20: Forecast metrics by asset and horizon

Horizon	Model	Set	Asset	MAE	RMSE	QLIKE [95% CI]
5	CNN-1D	Seq.	BBDC4	0.0140	0.0226	−5.1902 [−5.4841, −4.8206]
5	CNN-1D	Seq.	PETR4	0.0156	0.0240	−5.0598 [−5.2875, −4.7550]
5	CNN-1D	Seq.	VALE3	0.0134	0.0172	−5.3793 [−5.5121, −5.2431]
5	HistGradientBoosting	Set C C	BBDC4	0.0160	0.0239	−5.2507 [−5.4862, −4.9763]
5	HistGradientBoosting	Set C C	PETR4	0.0184	0.0264	−5.0532 [−5.2719, −4.7793]
5	HistGradientBoosting	Set C C	VALE3	0.0159	0.0199	−5.3646 [−5.4702, −5.2573]
5	MLP	Set A A	BBDC4	0.0135	0.0226	−5.1709 [−5.4817, −4.7816]
5	MLP	Set A A	PETR4	0.0154	0.0238	−5.0529 [−5.2731, −4.7620]
5	MLP	Set A A	VALE3	0.0126	0.0166	−5.4045 [−5.5466, −5.2630]
5	Random Forest	Set C C	BBDC4	0.0157	0.0234	−5.2852 [−5.5174, −5.0005]
5	Random Forest	Set C C	PETR4	0.0176	0.0259	−5.0806 [−5.2863, −4.7902]
5	Random Forest	Set C C	VALE3	0.0151	0.0187	−5.3866 [−5.4934, −5.2723]
5	Ridge	Set C C	BBDC4	0.0148	0.0232	−5.1685 [−5.4570, −4.8014]
5	Ridge	Set C C	PETR4	0.0157	0.0239	−5.0905 [−5.3091, −4.8093]
5	Ridge	Set C C	VALE3	0.0134	0.0170	−5.4319 [−5.5517, −5.3145]
20	CNN-1D	Seq.	BBDC4	0.0209	0.0325	−3.8624 [−4.1466, −3.4803]
20	CNN-1D	Seq.	PETR4	0.0228	0.0366	−3.6812 [−3.9561, −3.3330]
20	CNN-1D	Seq.	VALE3	0.0179	0.0227	−4.0220 [−4.1925, −3.8565]
20	HistGradientBoosting	Set C C	BBDC4	0.0237	0.0334	−3.9196 [−4.1527, −3.6527]
20	HistGradientBoosting	Set C C	PETR4	0.0296	0.0426	−3.6763 [−3.8730, −3.4413]
20	HistGradientBoosting	Set C C	VALE3	0.0262	0.0354	−3.9290 [−4.0547, −3.7984]
20	MLP	Set B B	BBDC4	0.0252	0.0355	−2.4923 [−3.6616, −0.6852]
20	MLP	Set B B	PETR4	0.0217	0.0338	−3.7469 [−3.9682, −3.4877]
20	MLP	Set B B	VALE3	0.0171	0.0223	−4.0420 [−4.2032, −3.8846]
20	Random Forest	Set C C	BBDC4	0.0262	0.0358	−3.8826 [−4.1009, −3.6177]
20	Random Forest	Set C C	PETR4	0.0275	0.0407	−3.7113 [−3.9222, −3.4600]
20	Random Forest	Set C C	VALE3	0.0254	0.0339	−3.9425 [−4.0739, −3.8066]
20	Ridge	Set C C	BBDC4	0.0237	0.0338	−3.8738 [−4.1222, −3.6050]
20	Ridge	Set C C	PETR4	0.0227	0.0350	−3.7405 [−3.9609, −3.4636]
20	Ridge	Set C C	VALE3	0.0194	0.0239	−4.0293 [−4.1752, −3.8752]

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The workflow uses locally processed B3/BovDBv2 daily adjusted price records. The public BovDBv2 dataset is available through Harvard Dataverse under DOI `10.7910/DVN/TMB4IG`. Derived scripts and reproducibility materials will be made available in a public repository upon publication, subject to redistribution constraints for raw market data.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT for language refinement, structural organization and code-review support. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Andrey V. S. Souza: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. **Bruno L. Dalmazo:** Methodology, Supervision, Writing - review and editing. **Giancarlo Lucca:** Methodology, Validation, Writing - review and editing. **Eduardo N. Borges:** Supervision, Methodology, Writing - review and editing. **Richard F. Pinto:** Software, Data curation, Validation. **Fabian C. Cardoso:** Data curation, Resources, Writing - review and editing. **Viviane L. D. de Mattos:** Formal analysis, Supervision, Writing - review and editing. **Rafael A. Berri:** Supervision, Project administration, Writing - review and editing.

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